Silver-rank zeros of the tensor-curl eta invariant on lens spaces: silver S-units, a Pell ladder, and a period law

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Abstract

Let $\eta_J(L(p,q))$ be the eta invariant of the rank-J projected transverse-traceless tensor-curl operator on the round lens space $L(p,q)$ [1]. At rank one it is $-4s(q,p)$, which vanishes exactly on the amphicheiral spaces $q^2 \equiv -1 \pmod p$, so there a zero reads as mirror symmetry. At higher rank this fails on the silver levels $p \mid 2J^2-1$, where η_J is an even integer and zeros of chiral spaces are abundant. We study the zero set at the level $p = 2J^2-1$. Reading $\mathbb{Z}[\sqrt{2}]$ modulo p through $\sqrt{2} \mapsto 2J$, the four zero families of [1] become the orbits of the silver S-units $\sqrt{2}$, $\delta/\sqrt{2}$, δ and 2δ, $\delta = 1+\sqrt{2}$. We prove a fifth family, $\eta_J(L(p,3-4J)) = J-1$ for odd J and 0 for even J, and an infinite Pell ladder on the orbits of the powers δ^n: with P_n the Pell numbers, $\eta_J = \frac{2}{3}P_{n-2}J$ for even n and 0 for odd n when $P_n \mid J$, and, for $n \geq 2$, $\eta_J = \gamma_n(J-1)/P_n$ with an explicit $\gamma_n > 0$ when $J \equiv 1 \pmod{P_n}$. We conjecture a period law (every nonconstant S-unit orbit is eventually quasi-polynomial in J with period dividing its coefficient of $\sqrt{2}$), support it by a reported out-of-sample test on 64 orbits, and prove it for every nonconstant orbit of norm ±1 or ±2 (the powers δ^n, $n \geq 1$, and the norm-two orbits; the constant orbit $q = \pm1$ lies outside the law and is cubic in J). There, on each residue class of J, η_J is a linear function of J past an explicit threshold; the zero set of each orbit is computable by finitely many exact evaluations and is eventually periodic, with density equal to the proportion of zero classes. Among these orbits exactly the first three families of [1] vanish at every rank, and in a window of 64 orbits exactly the four families do. Consequently, for every K there is a rank at which K pairwise non-homeomorphic chiral lens spaces, none balanced level by level, all have $\eta_J = 0$. The period law beyond norm ±2, the zeros outside the S-units and the complete zero set remain open.

1 Introduction

The paper [1] evaluates the eta invariant $\eta_J(L(p,q))$ of the rank-J projected transverse-traceless (TT) tensor-curl operator on every round lens space and exhibits four infinite families of zeros at the level $p = 2J^2-1$: $q = J$, $J+1$, $2J-1$ and $J(J-1)$. The eta invariant measures the net helicity of the spectrum, and a natural first guess is that it vanishes only for a reason of symmetry. At rank one that guess is right, by composition of classical results: $\eta_1(L(p,q)) = -4s(q,p)$ [1], the Dedekind sum $s(q,p)$ vanishes iff $q^2 \equiv -1 \pmod p$ [6, p. 36], and $q^2 \equiv -1$ is the criterion for $L(p,q)$ to be amphicheiral [5, 4]; see [2, Cor. 4.4]. The companion paper [2] shows that at prime moduli $p \geq 5$ the same dichotomy persists at every rank J with $2J^2 \not\equiv 1 \pmod p$ and $p \nmid J$ (Corollary 2.6(c) below). The exceptions are the *silver ranks* $2J^2 \equiv 1 \pmod p$. There η_J is an even integer, and zeros of

chiral spaces are common: at the 31 silver primes in $[211, 599]$ about 15% of the q-orbits are zeros [2, Obs. 5.6]. This paper is about those zeros, at the level $p = 2J^2 - 1$ itself.

Results.

(1) *A coordinate.* The ring homomorphism $\mathbb{Z}[\sqrt{2}] \to \mathbb{Z}/p$, $\sqrt{2} \mapsto 2J$, sends the silver S-units $\pm\delta^n\sqrt{2}^m$ ($\delta = 1 + \sqrt{2}$) to units, and the four families of [1] are the orbits of $\sqrt{2}$, $\delta/\sqrt{2}$, δ and 2δ (Lemma 4.1). The Cayley coordinate $\rho = (1 + q)/(1 - q)$ gives a dictionary for the known zero orbits (Proposition 4.3).

(2) *A fifth family* (Theorem 5.1): $\eta_J(L(2J^2 - 1, 3 - 4J)) = (J - 1)\mathbf{1}_{J \text{ odd}}$, the orbit of δ^{-2}.

(3) *The Pell ladder* (Theorem 6.1): exact values of η_J on the orbit of δ^n on the two slices $P_n \mid J$ and $J \equiv 1 \pmod{P_n}$. The first two rungs are the third family of [1] and the fifth family; the orbit of δ^n vanishes at every rank only for $n = 1$.

(4) *The period law.* Conjecture 7.1 with its out-of-sample evidence, and Theorem 8.4, which proves it for every nonconstant orbit of norm ± 1 or ± 2 (the constant orbit $q = \pm 1$ is outside the law and cubic in J), with thresholds, linear class polynomials, a decision procedure and complete zero sets for 24 families, including one isolated zero ($J = 429$) that is not part of any zero class. The zero sets are eventually periodic, with an explicit density (Corollary 8.5).

(5) *Rank-uniform zeros.* Among the orbits of norm ± 1 and ± 2, exactly the first three families of [1] vanish at every rank (Corollary 8.9); in a window of 64 orbits, exactly the four families do (Corollary 7.2(a)).

(6) *Unbounded ambiguity* (Corollary 8.13): for every K there is a rank at which K pairwise non-homeomorphic chiral lens spaces, none balanced level by level, all have $\eta_J = 0$. This is the precise sense in which the rank-one dictionary "zero = mirror" fails at higher rank: a single vanishing eta invariant can hide an unbounded amount of handedness.

Along the way we record a refuted conjecture (Conjecture 4.6) and a refuted "zero-term principle" (Remark 7.4), and explain why the conjugacy mechanism that proves the family $q = J$ at every divisor of $2J^2 - 1$ in [2] does not close the classification (Remark 4.5).

What is proved, what is observed. Every statement is labelled as proved, conjectural or observed. Observations are finite computations whose ranges are stated. The Faulhaber-sum evaluations and the leading-coefficient table are certificate-assisted (Remarks 6.4 and 8.11, checked by two independent computer algebra routes). The finite threshold and class-polynomial reconstruction also uses exact algebraic checks, as specified in Theorem 8.4. Exact computations are listed in Appendix A; they test the identities and conventions, and do not replace the proofs. No physical claim is made. The methods are elementary (Dedekind reciprocity, Pell arithmetic, lattice counting and an integrality argument), and Dedekind sums at Pell-type arguments are classical territory; we have not made a literature search for the individual evaluations and claim no priority for them. Eventual quasi-polynomial behaviour of parametric counting functions is also classical in general; for context see the parametric Presburger arithmetic of Bogart, Goodrick and Woods [3]. We do not claim quasi-polynomiality as such. What is specific here is the explicit decomposition of η_J into a Dedekind term, a residue term and a lattice box, the cancellation that makes every class linear, the explicit thresholds, and the resulting zero-set procedure in these sectors.

Relation to the companion papers. We use the operator, the normalization and the master formula of [1], and the following results of the companion [2], restated in Section 2: the classification of level-wise balance, the Dedekind-sum facts, the universal congruence $\eta_J - J(2J^2-1)\eta_1 \in 2\mathbb{Z}$, the exact denominator and the prime-modulus dichotomy, the zero $\eta_J(L(p,J)) = 0$ at every divisor $p > 1$ of $2J^2-1$, and the Dedekind sum at the silver unit. No result of [2] depends on the present paper; its Section 9 summarizes the present results.

Structure of the proofs. Section 4 introduces the two coordinates. The fifth-family and Pell-ladder proofs evaluate the Dedekind, residue and lattice terms separately. Section 7 states the period-law conjecture and its finite evidence. Section 8 proves the law in the norm-one and norm-two sectors, using an integrality argument and linear breakpoint bounds, and derives the zero-set procedure and unbounded ambiguity.

2 Conventions and imported results

We keep the conventions of [1, Sec. 2] and [2, Sec. 2]. For $p \geq 2$ and $q \in (\mathbb{Z}/p\mathbb{Z})^\times$ the generator of $\mathbb{Z}_p$ acts on $S^3 \subset \mathbb{C}^2$ by $(z_1, z_2) \mapsto (\mathrm{e}^{2\pi i/p} z_1, \mathrm{e}^{2\pi i q/p} z_2)$, and $L(p,q)$ is the quotient with its complex quotient orientation. We write q^* for an inverse of q modulo p, $\langle k \rangle_p$ for the least non-negative residue, and $\tilde{q}$ for the odd lift of q ($\tilde{q} = q$ if q is odd, $\tilde{q} = q + p$ if q is even and p odd). Put

$$A = \frac{1+\tilde{q}}{2}, \qquad B = \frac{1-\tilde{q}}{2}, \qquad A + B = 1. \tag{1}$$

Let $W_a = \{-a, -a+2, \dots, a\}$ be the weights of the $SU(2)$ representation V_a. The lifted generator acts on the weight vector (w_L, w_R) of $V_a \otimes V_b$ by $\exp(2\pi i(Aw_L + Bw_R)/p)$, and for $a \equiv b \pmod 2$ the invariant count is

$$I_{a+1,b+1}(p,q) = \#\{(w_L, w_R) \in W_a \times W_b : \; Aw_L + Bw_R \equiv 0 \pmod p\}.$$

The rank-J operator has signed spectrum $+(n+J+1)$ on $V_{n+2J} \otimes V_n$ and $-(n+J+1)$ on $V_n \otimes V_{n+2J}$, $n \geq 0$ [1, Thm. 3.1]. With $D_J(n) = I_{n+2J+1,n+1} - I_{n+1,n+2J+1}$ the level imbalance, η_J is the value at $s = 0$ of the continuation of $\sum_{n\geq 0} D_J(n)(n+J+1)^{-s}$ [1, Prop. 4.1]. We use the lens symmetries

$$\eta_J(L(p,q^*)) = \eta_J(L(p,q)), \qquad \eta_J(L(p,-q)) = -\eta_J(L(p,q)), \tag{2}$$

the Dedekind sum $s(q,p) = \sum_{a=0}^{p-1} ((a/p))\,((qa/p))$, the rank-one value $\eta_1(L(p,q)) = -4s(q,p)$ [1, Cor. 5.8], and the master formula [1, Thm. 5.1]

$$\begin{aligned}\eta_J(L(p,q)) = \; & -4J\, s(q,p) + \frac{2}{p}\sum_{a=1}^{J-1}(J-a)\,[\rho_p(aq) + \rho_p(aq^*)] \\ & - 2\sum_{\substack{a,b\geq 1 \\ a+b\leq J-1}} (J-a-b)\,[\delta_p(aq-b) - \delta_p(aq+b)] - L_J(p,q),\end{aligned} \tag{3}$$

with $\rho_p(k) = 0$ if $p \mid k$ and $p - 2\langle k \rangle_p$ otherwise, $\delta_p(k) = \mathbf{1}_{p|k}$, and the low-mode term $L_J(p,q) = \sum_{m=1}^{J-1}[I_{J-m,J+m}(p,q) - I_{J+m,J-m}(p,q)]$. Equation (3) holds for every $p \geq 2$, every unit q and every $J \geq 1$.

Definition 2.1. Let q be a unit modulo p and $J \geq 1$. $L(p,q)$ is *rank-J balanced* if $D_J(n) = 0$ for every level $n \geq 0$; it is *mirror* (amphicheiral) if $q^2 \equiv -1 \pmod p$ and *chiral* otherwise [5, 4]. The rank J is a *silver rank* for p if $2J^2 \equiv 1 \pmod p$, and a *blind rank* if $p \mid 2J$. The *Clifford condition* is $\gcd(q \mp 1, p) \leq 2$.

The following results of [2] are used; we restate them with their numbering there.

Theorem 2.2 ([2, Thm. 4.1]). *Let $p \geq 3$, q a unit and $J \geq 1$. $L(p,q)$ is rank-J balanced iff* (a) $q^2 \equiv -1 \pmod p$*; or* (b) $p \mid 4J^2 - 2$ *and* $q \equiv \pm(2J \pm 1) \pmod p$*; or* (c) $p \mid 2J$ *and the Clifford condition holds.*

Lemma 2.3 ([2, Lem. 6.2]). *For $p \geq 2$ and q a unit, $T := 12p\, s(q,p) \in 2\mathbb{Z}$; in particular $6p\, s(q,p) \in \mathbb{Z}$.*

Theorem 2.4 (Universal congruence, [2, Thm. 6.7]). *For every $p \geq 2$, unit q and $J \geq 1$, $\eta_J(L(p,q)) - J(2J^2-1)\,\eta_1(L(p,q)) \in 2\mathbb{Z}$.*

Theorem 2.5 ([2, Thm. 6.8(c)]). *Write $p = 2^a 3^b m$ with $\gcd(m,6) = 1$ and $C = J(2J^2-1)$. Then $\eta_J(L(p,q)) \in 2\mathbb{Z}$ iff $m \mid C(q^2+1)$, ($a < 2$ or $2^a \mid J$), and ($b = 0$ or $3^{b+1} \mid J$).*

Corollary 2.6 ([2, Cor. 6.4]). *Let $p \geq 5$ be prime.* (b) *If $2J^2 \equiv 1 \pmod p$ then $\eta_J(L(p,q)) \in 2\mathbb{Z}$ for every unit q.* (c) *If $2J^2 \not\equiv 1 \pmod p$ and $p \nmid J$, then $\eta_J(L(p,q)) = 0$ iff $q^2 \equiv -1 \pmod p$.*

For $J \geq 2$ and a divisor $p > 1$ of $2J^2 - 1$ put

$$\varepsilon := 4J + 3, \qquad \varepsilon^2 - 6\varepsilon + 1 = 8(2J^2 - 1) \equiv 0 \pmod p, \tag{4}$$

the silver unit $3 + 2\sqrt{2}$ read modulo p with $\sqrt{2} \equiv 2J$.

Theorem 2.7 ([2, Thm. 7.1]). *For every $J \geq 2$ and every divisor $p > 1$ of $2J^2 - 1$, $\eta_J(L(p,J)) = 0$, and hence $\eta_J(L(p,q)) = 0$ for $q \equiv \pm J$ and $q \equiv \pm 2J \pmod p$.*

Corollary 2.8 ([2, Cor. 7.11]). *Let p be odd and $\varepsilon^2 - 6\varepsilon + 1 \equiv 0 \pmod p$. Then $s(\varepsilon,p) = -(p-1)/(2p)$ if $p \equiv 1 \pmod 8$ and $1/(2p)$ if $p \equiv 7 \pmod 8$.*

3 Zeros at silver ranks: what is known

- (Proved.) At a silver rank of a prime $p \geq 5$, $\eta_J(L(p,q))$ is an even integer for every unit q (Corollary 2.6(b)); a silver rank exists iff $p \equiv \pm 1 \pmod 8$. At composite $p \mid 2J^2 - 1$ (odd, coprime to 3), $\eta_J \in 2\mathbb{Z}$ iff $p \mid J(2J^2-1)(q^2+1)$, which holds (Theorem 2.5), so η_J is an even integer at every silver rank of every modulus.
- (Proved zero orbits.) The balanced family $q \equiv \pm(2J \pm 1)$ (Theorem 2.2(b)); the orbit $q \equiv \pm J, \pm 2J$ (Theorem 2.7); the mirror orbit when -1 is a square modulo p; and, when $p = 2J^2 - 1$ exactly, the orbits of $q = J + 1$ and $q = J(J-1)$ of [1, Thm. 7.1], the fifth family of Theorem 5.1 (even J), and the Pell ladder of Theorem 6.1.
- (Observed.) [2, Obs. 5.5 and 5.6]: at the 31 silver primes in $[211, 599]$, 471 of 3,076 q-orbits (15.3%) are zeros, with the zero fraction highest at small silver ranks (33% at $p = 241$, $J = 11$); at $(97,7)$ and $(199,10)$ the cube roots of unity are zeros. In the divisor family of Theorem 2.7 with $J \leq 59$, further zeros occur at roughly 25–30% of the rows examined in the supplement [7]. These fractions are descriptive statistics of the examined ranges, computed with different orbit conventions, and are not asserted to have limits.

4 The silver S-unit coordinate and the Cayley coordinate

From here on $p = 2J^2 - 1$ unless a divisor is explicitly allowed, and $\mathbb{Z}[\sqrt{2}]$ is read modulo p through

$$\sqrt{2} \longmapsto 2J, \qquad \text{since } (2J)^2 = 2p + 2 \equiv 2 \pmod{p}. \tag{5}$$

This is a ring homomorphism $\mathbb{Z}[\sqrt{2}] \to \mathbb{Z}/p$. It sends $\delta := 1 + \sqrt{2}$ to $2J + 1$, $\delta^{-1} = \sqrt{2} - 1$ to $2J - 1$, $\delta^2 = 3 + 2\sqrt{2}$ to $4J + 3 = \varepsilon$ (the silver unit of (4)), $\delta^{-2} = 3 - 2\sqrt{2}$ to $3 - 4J$, and 2 to a unit, p being odd. Hence every *silver S-unit* $\pm\delta^n\sqrt{2}^m$ $(n, m \in \mathbb{Z})$ reduces to a unit modulo p; the same holds modulo every divisor $p > 1$ of $2J^2 - 1$, where (5) still holds. For $\xi = x + y\sqrt{2} \in \mathbb{Z}[\sqrt{2}]$ the reduction is $\xi(J) = x + 2yJ$. Since η_J depends on q only through the lens orbit $\{\pm q^{\pm 1}\}$ up to sign (2), we speak of the *orbit* of an S-unit ξ, meaning $\{\pm\xi^{\pm 1}\}$, and write (n, m) for the orbit of $\delta^n\sqrt{2}^m$; every orbit has a representative in $\mathbb{Z}[\sqrt{2}]$ (replace ξ by ξ^{-1} if necessary).

Lemma 4.1 (The four known families as S-unit orbits)**.** *Let $J \geq 2$ and $p = 2J^2 - 1$. Modulo p,*

$$J \equiv \sqrt{2}^{-1}, \qquad J + 1 \equiv \delta/\sqrt{2}, \qquad 2J - 1 \equiv \delta^{-1}, \qquad J(J - 1) \equiv -(2\delta)^{-1}.$$

Thus the four zero families $q = J$, $J + 1$, $2J - 1$, $J(J - 1)$ of [1, Thm. 7.1] are the orbits $(0, 1)$, $(1, -1)$, $(1, 0)$, $(1, 2)$ of $\sqrt{2}$, $\delta/\sqrt{2}$, δ and 2δ. The same congruences hold modulo every divisor $p > 1$ of $2J^2 - 1$.

Proof. $2J \cdot J = 2J^2 \equiv 1$, so $J \equiv (2J)^{-1} \equiv \sqrt{2}^{-1}$; then $J + 1 \equiv \sqrt{2}^{-1} + 1 = (1 + \sqrt{2})/\sqrt{2}$; $2J - 1 \equiv \sqrt{2} - 1 = \delta^{-1}$; and $(4J + 2)\,J(J - 1) = 2J(2J^2 - 1) - 2J^2 \equiv -1$, so $J(J - 1) \equiv -(4J + 2)^{-1} = -(2 + 2\sqrt{2})^{-1}$. Only $2J^2 \equiv 1$ was used. □

Definition 4.2 (Cayley coordinate)**.** Let p be odd and

$$D_p = \{q \in (\mathbb{Z}/p)^\times : \ \gcd(1 - q, p) = \gcd(1 + q, p) = 1\} = \{q \in (\mathbb{Z}/p)^\times : \ \gcd(q^2 - 1, p) = 1\}.$$

For $q \in D_p$ put

$$\rho = \rho(q) = \frac{1 + q}{1 - q} \in (\mathbb{Z}/p)^\times, \tag{6}$$

the ratio A/B of the two coefficients in (1), i.e. the ratio of the left and right Clifford rotation angles of the generator of $L(p, q)$.

At a prime p, D_p is the set of units $q \not\equiv \pm 1$. At a composite p it is smaller, and outside it ρ is not a unit: for $J = 5$, $p = 49$, the unit $q = 8$ has $1 - q = -7$ not invertible, and $q = 6$ gives $1 + q = 7$ and $\rho = 28$. On D_p the map $q \mapsto \rho$ is a Möbius transformation, hence injective, with inverse $q = (\rho - 1)/(\rho + 1)$, and $\rho \not\equiv \pm 1$. The set D_p is closed under $q \mapsto q^{-1}$ (as $1 - q^{-1} = -q^{-1}(1 - q)$) and $q \mapsto -q$ (which exchanges $1 \pm q$); $q \mapsto q^{-1}$ sends $\rho \mapsto -\rho$ and $q \mapsto -q$ sends $\rho \mapsto \rho^{-1}$, so the lens orbit $\{\pm q^{\pm 1}\}$ corresponds to $\{\pm\rho^{\pm 1}\}$, and orientation reversal $(\eta_J \mapsto -\eta_J)$ is $\rho \mapsto \rho^{-1}$.

Proposition 4.3 (Dictionary at silver ranks)**.** *Let $p > 1$ divide $2J^2 - 1$ and read $\sqrt{2} \equiv 2J$, $\delta \equiv 2J + 1$ as above. The orbits below lie in D_p (the mirror orbit whenever it exists), and in the coordinate ρ:*

(a) the balanced family (b) of Theorem 2.2, $q \equiv \pm(2J \pm 1)$, is $\rho \in \{\pm\delta^{\pm 1}\}$;

(b) the orbit of Theorem 2.7, $q \equiv \pm J^{\pm 1}$, is $\rho \in \{\pm\delta^{\pm 2}\}$;

(c) the mirror orbit $q^2 \equiv -1$ is $\rho^2 \equiv -1$;

(d) the orbit of $q \equiv 3 - 4J \equiv \delta^{-2}$ *is* $\rho \in \{\pm\sqrt{2}^{\pm 1}\}$*, with* $\rho(3 - 4J) = \sqrt{2}$ *exactly.*

Proof. Only $2J^2 \equiv 1$ is used. *Membership in* D_p. The residues $2J$, $J \pm 1$ and $2J \pm 1$ are units: a prime dividing $J \pm 1$ and $2J^2 - 1$ divides $2J^2 - 1 - 2(J \mp 1)(J \pm 1) = 1$, and $(2J+1)(2J-1) = 2(2J^2-1)+1$. For $q = \pm(2J \pm 1)$ the numbers $1 \pm q$ are, up to sign, among $2J$, $2(J+1)$, $2(J-1)$; for $q = \pm J$, $\pm 2J$ they are $1 \pm J$, $1 \pm 2J$; for $q = 3 - 4J$ they are $2(2J-1)$ and $-4(J-1)$, and for its inverse $3 + 4J$ they are $-2(2J+1)$ and $4(J+1)$; and for $q^2 \equiv -1$, $q^2 - 1 \equiv -2$. All are units, p being odd, and D_p is closed under $q \mapsto \pm q^{\pm 1}$. (A check over every divisor $p > 1$ of $2J^2 - 1$, $2 \le J \le 100$, found no exception [7].) (a) For $q = 2J + 1$: $\rho = (2J+2)/(-2J) = -(J+1)J^{-1} = -(J+1) \cdot 2J = -(2J^2 + 2J) \equiv -(2J+1) = -\delta$. (b) For $q = J$: $(3 + 4J)(1 - J) = 3 + J - 4J^2 \equiv 1 + J$, so $\rho = (1+J)/(1-J) = 3 + 4J = \delta^2$. (c) $(1+q)^2 + (1-q)^2 = 2(1+q^2)$, so $\rho^2 \equiv -1$ iff $q^2 \equiv -1$. (d) For $q = 3 - 4J$: $\rho = (4 - 4J)/(4J - 2) = (2 - 2J)/(2J - 1)$, and $2J(2J-1) = 4J^2 - 2J \equiv 2 - 2J$, so $\rho = 2J = \sqrt{2}$. The orbit statements follow from the transformation rules above. □

Observation 4.4 (Silver primes in the coordinate ρ)**.** The following was found in exploratory computations, every prime $p \le 673$ that has a silver rank being taken at its minimal silver rank J (the smaller root of $2J^2 \equiv 1$). "Sporadic" means: not in the three orbits of (a).

(a) (Universal zero orbits.) The orbits that are zeros at every silver prime examined are the two S-unit orbits of Proposition 4.3(a),(b), namely family (b), witnessed by level balance, and the orbit of Theorem 2.7, witnessed by the inert class R^2L^2 of [2, Thm. 8.2(viii)]; and the mirror orbit, witnessed by orientation reversal, which is not an S-unit orbit and exists exactly at the silver primes $p \equiv 1 \pmod 8$. Among the S-units $q = s \cdot 2^c \sqrt{2}^{\,a} \delta^b$ with $s = \pm 1$, $|c| \le 2$, $|a| \le 3$, $|b| \le 4$, exactly the 28 expressions lying in those two orbits are zeros at all 57 silver primes $17 \le p \le 673$; among $\rho = \pm\delta^k$, $1 \le k \le 14$, exactly $k = 1, 2$ are (the next best, $k = 14$, at 16 of 56 primes where it is defined). These searches were first run during exploration on smaller prime ranges. With the expression ranges recovered from the retained scripts, repetition over the stated prime range tested and confirmed the same predicted universal-zero sets [7].

(b) (Sporadic zeros, primes < 400.) The sporadic zeros show no structure in ρ on the unit group or in lattice size: on the subgroup $\langle -1, \delta \rangle$ they occur at the subgroup's density (for instance 0 of 3 at $p = 41$ and 0 of 12 at $p = 199$); their rate is flat, 16–20%, across height buckets of the $\mathbb{Z}[\sqrt{2}]$-lattice; and the hypothesis that the cofactor norm lies in $\{m, 2m\}$ is refuted (rates 5/51 and 8/45 against 229/1384 for all other classes).

(c) (One signal.) Among sporadic cases, lifts of norm 2 ($\rho \equiv \sqrt{2} \cdot$unit) are zeros at $40/137 = 29\%$ against 15% otherwise (primes ≤ 400); the excess sits at $\rho = \pm\sqrt{2}$ exactly, i.e. at $q \equiv 3 - 4J \equiv 2(J-1)^2$. At $\rho = \sqrt{2}$ the zero occurs at 0 of the 28 primes $p \equiv 1 \pmod 8$ and at 18 of the 29 primes $p \equiv 7 \pmod 8$ with $p \le 673$; and at the primes $p = 2J^2 - 1$ with $3 \le J \le 18$ every row fits $\eta_J(L(p, 3 - 4J)) = J - 1$ for odd J and 0 for even J.

Remark 4.5 (Why the mechanism of [2, Sec. 7] does not extend)**.** The proof of Theorem 2.7 in [2, Sec. 7] succeeded because at $q = J$ the summand of the trace form [2, eq. (19)] collapsed [2, Lems. 7.2–7.3] to a single two-factor Fourier–Dedekind sum with the quarter shift of [2, Lem. 7.4], and that shift produced one untwisted Rademacher symbol. For general q this does not happen. In the variable $\zeta = \mathrm{e}^{2\pi i k/p}$ the denominators of the two terms of the trace form are, up to constants and roots of unity, $(\zeta^a - 1)(\zeta^{q-1} - 1)(\zeta^{q+1} - 1)$ with $a \in \{1, q\}$, so $p\,\eta_J$ is a combination of three-factor Fourier–Dedekind sums. With $A = \zeta^{q-1} - 1$, $B = \zeta^{q+1} - 1$ and $C = \zeta^2 - 1$ one has $B - A = \zeta^{q-1} C$, hence the cocycle identity

$$\frac{1}{AB} = \frac{1}{C}\Big(\frac{1}{A} - \frac{1}{B}\Big) - \frac{1}{B},$$

which reduces them to two-factor sums; but the resulting shifts are Jq and $(J+1)q$, not quarter shifts, and the sums are twisted Dedekind sums (Rademacher symbols at congruence level) rather than one untwisted Ψ. Thus the conjugacy mechanism does not by itself close the classification at silver ranks. Nothing about the twisted sums is claimed here.

Conjecture 4.6 (Nonvanishing at $q = 3 - 4J$ for primes $p \equiv 1 \pmod 8$; refuted)**.** *For every prime $p \equiv 1 \pmod 8$ with minimal silver rank J, $\eta_J(L(p, 3-4J)) \neq 0$.*

This was formulated, together with Observation 4.4(c), before the 28 primes $p \equiv 1 \pmod 8$ with $761 \le p \le 1553$ were computed. It is false: 27 of them give $\eta_J \neq 0$, but $p = 1129$, $J = 203$ gives $\eta_{203}(L(1129,\, 3 - 4 \cdot 203)) = 0$. Together with the exploratory prime sample, the zero at $\rho = \sqrt{2}$ occurs at 1 of 56 primes $p \equiv 1 \pmod 8$ and at 18 of 29 primes $p \equiv 7 \pmod 8$; the split by $p \bmod 8$ is a statistic, not a law, and no replacement conjecture is made. (The complementary control "a zero at every silver prime" fails at $p \equiv 1 \pmod 8$.)

5 The fifth family at $p = 2J^2 - 1$

The orbit $\rho = \sqrt{2}$, i.e. $q \equiv 3 - 4J \equiv \delta^{-2}$, is the one place where Observation 4.4 showed structure. Its value at $p = 2J^2 - 1$ was conjectured from the exploratory rows $3 \le J \le 18$ with p prime, confirmed on every $J \in [19, 60]$ with p prime in a subsequent test of that prediction (19 rows, Appendix A), and then proved for every J, prime or composite p, by the four-term method of [1, Thm. 7.1]. Write the master formula (3) as

$$\eta_J(L(p,q)) = -4J\, s(q,p) + R_J(p,q) - 2\Delta_J(p,q) - L_J(p,q), \tag{7}$$

with the residue term and the triangular sum

$$R_J = \frac{2}{p}\sum_{a=1}^{J-1}(J-a)\big[\rho_p(aq) + \rho_p(aq^*)\big], \qquad \Delta_J = \sum_{\substack{a,b\ge 1\\ a+b\le J-1}} (J-a-b)\big[\delta_p(aq-b) - \delta_p(aq+b)\big],$$

and the low-mode term L_J of Section 2. Throughout, q^* is an integer inverse of q modulo p, $\mathbf{1}_{\mathcal{P}} \in \{0,1\}$ is the indicator of a proposition $\mathcal{P}$, and

$$\mathcal{T}(x,y) = \frac{1}{12}\Big(\frac{x}{y} + \frac{y}{x} + \frac{1}{xy}\Big) - \frac{1}{4}, \qquad s(x,y) + s(y,x) = \mathcal{T}(x,y) \quad (x,y \ge 1,\ \gcd(x,y) = 1), \tag{8}$$

is Dedekind reciprocity.

Theorem 5.1 (The fifth family)**.** *Let $J \ge 2$, $p = 2J^2 - 1$ (prime or not) and $q \equiv 3 - 4J \pmod p$, whose least positive residue is $2(J-1)^2$ and whose inverse is $q^* \equiv 3 + 4J$. Then*

$$\eta_J(L(p,q)) = (J-1)\,\mathbf{1}_{J \text{ odd}}. \tag{9}$$

In particular $q \equiv 3 - 4J$ is a zero of η_J exactly for even J, and the same value holds on the orbit $\pm q^{\pm 1}$ up to the sign of q.

Proof. Units. $(3-4J)(3+4J) = 9 - 16J^2 = 1 - 8p$, so $q^* \equiv 3 + 4J =: h$; and $8p = (4J+3)(4J-3)+1$ gives $\gcd(h,p) = 1$.

Step 1 (the Dedekind sum). $s(q,p) = s(q^*,p) = s(h,p)$. By (8), $s(h,p) = -s(p,h) + \mathcal{T}(h,p)$. Since $8p \equiv 1 \pmod h$, $s(p,h) = s(8,h)$, and reciprocity again gives $s(8,h) = -s(h,8) + \mathcal{T}(8,h)$. Using $h^2 + 1 - 8p = 6h$ the two reciprocity terms collapse to $\mathcal{T}(h,p) - \mathcal{T}(8,h) = \frac{1}{2p} - \frac{1}{16}$, so

$$s(h,p) = s(h,8) + \frac{1}{2p} - \frac{1}{16}.$$

Here $h \equiv 3 \pmod 8$ for even J and $h \equiv 7 \pmod 8$ for odd J, and $s(3,8) = \frac{1}{16}$, $s(7,8) = -\frac{7}{16}$. Hence

$$s(q,p) = \frac{1}{2p} - \frac{\mathbf{1}_{J \text{ odd}}}{2}, \tag{10}$$

which is the value of Corollary 2.8 at $\varepsilon = q^*$, since $p \equiv 7 \pmod 8$ for even J and $p \equiv 1 \pmod 8$ for odd J.

Step 2 (the residue term). For $1 \le a \le J-1$ both $a(4J+3)$ and $a(4J-3)$ are less than $2p$, so each residue wraps at most once; the wrap thresholds are

$$\frac{p}{4J+3} = \frac{J}{2} - \frac{3}{8} + \frac{1}{8(4J+3)}, \qquad \frac{p}{4J-3} = \frac{J}{2} + \frac{3}{8} + \frac{1}{8(4J-3)}.$$

With $aq \equiv -a(4J-3)$, $aq^* = a(4J+3)$, $\rho_p(-k) = -\rho_p(k)$ and $\rho_p(k) = p - 2\langle k \rangle_p$, an index a at which neither or both residues wrap contributes $\rho_p(aq) + \rho_p(aq^*) = -12a$; the only mixed index is $a = J/2$ for even J, where aq^* wraps and $a(4J-3)$ does not, and it contributes $2p - 12a$. For odd J there is no mixed index. Therefore

$$R_J = -\frac{4J(J^2-1)}{p} + 2J\,\mathbf{1}_{J \text{ even}}. \tag{11}$$

Step 3 (the triangular sum vanishes). A term $aq \pm b \equiv 0$ with $a, b \ge 1$, $a + b \le J-1$ means $a(4J-3) \mp b \in \{p, 2p\}$. Each case confines a to an interval of length less than $\frac{1}{2}$ containing no integer, or forces $a > J-1$: $a(4J-3) + b = p$ needs $\frac{J}{2} + \frac{J}{4(J-1)} \le a < \frac{J}{2} + \frac{3}{8}$; $a(4J-3) - b = p$ needs $\frac{J}{2} + \frac{3}{8} < a \le \frac{J+1}{2} - \frac{1}{2(2J-1)}$; and the value $2p$ needs $a > J-1$. Hence $\Delta_J = 0$ (for $J = 2$ the sum is empty).

Step 4 (the low-mode term). As in Section 2, a weight pair $(w_L, w_R) \in W_{a-1} \times W_{b-1}$ is invariant iff $(1+\tilde{q})w_L + (1-\tilde{q})w_R \equiv 0 \pmod{2p}$; with $u = (w_L - w_R)/2$, $v = (w_L + w_R)/2 \in \mathbb{Z}$ (as $w_L \equiv w_R \pmod 2$) this reads $v + qu \equiv 0 \pmod p$, i.e. $v \equiv (4J-3)u$, and in the low-mode range $|u|, |v| \le J-1$. Writing $v = (4J-3)u - \ell p$: $\ell = 0$ forces $u = v = 0$; $\ell = 1$ needs $(4J-3)u \in [p - J + 1, p + J - 1]$, i.e. $u \in \left[\frac{J}{2} + \frac{J/2}{4J-3},\ \frac{J}{2} + \frac{5J/2-2}{4J-3}\right]$, which contains an integer only for odd J, namely $u = \frac{J+1}{2}$, $v = \frac{J-1}{2}$, i.e. $(w_L, w_R) = (J, -1)$; $\ell = 2$ would need $u > J-1$; negative ℓ give the negatives of these points. The solution $(0,0)$ lies in $W_{a-1} \times W_{b-1}$ iff a and b are both odd, and $J \pm m$ have the same parity, so it contributes equally to $I_{J-m,J+m}$ and $I_{J+m,J-m}$ and cancels in L_J. The solutions $\pm(J,-1)$ never lie in $W_{J-m-1} \times W_{J+m-1}$ (as $|w_L| = J > J - m - 1$), and they lie in $W_{J+m-1} \times W_{J-m-1}$ exactly when m is odd and $1 \le m \le J-2$ (the parity and size conditions of the two weight sets), which requires J odd. There are $\frac{J-1}{2}$ such m, each contributing 2 to $I_{J+m,J-m}$, so

$$L_J = -(J-1)\,\mathbf{1}_{J \text{ odd}}. \tag{12}$$

Assembly. By (7) and (10)–(12),

$$\begin{aligned}\eta_J &= -4J\Big(\frac{1}{2p} - \frac{\mathbf{1}_{J \text{ odd}}}{2}\Big) - \frac{4J(J^2-1)}{p} + 2J\,\mathbf{1}_{J \text{ even}} + (J-1)\,\mathbf{1}_{J \text{ odd}} \\ &= -\frac{2J}{p}\big(1 + 2(J^2-1)\big) + 2J + (J-1)\,\mathbf{1}_{J \text{ odd}},\end{aligned}$$

and $1 + 2(J^2-1) = p$, which gives (9). The orbit statement is (2). □

Remark 5.2. (i) Formula (9) was first conjectured from the exploratory rows $3 \le J \le 18$ with p prime (for instance $\eta_J(L(p, 3-4J)) = 2, 0, 6, 0$ at $p = 17, 31, 97, 127$) and predicted the value on the 19 rows $J \in \{21, 22, 24, 25, 28, 34, 36, 38, 39, 41, 42, 43, 45, 46, 49, 50, 52, 56, 59\}$ (p prime) before they were computed: the odd J gave $20, 24, 38, 40, 42, 44, 48, 58$ and every even J gave 0. The proof adds the composite levels, on which nothing had been computed; for instance $J = 5$, $p = 49$, $\eta_5 = 4$; $J = 9$, $p = 161$, $\eta_9 = 8$; $J = 12$, $p = 287$, $\eta_{12} = 0$. (ii) On every row of the prime-level verification ($J \le 60$) with $J \ge 6$, the orbit of $3 - 4J$ is distinct from the four orbits of Lemma 4.1, from family (b), from the orbit of Theorem 2.7 and from the mirror orbit; at $J = 4$ it coincides with the orbit of $J(J-1)$ and at $J = 2$ with every family. On the rows examined it is therefore a fifth family at $p = 2J^2 - 1$ for $J \ge 6$; a proof of distinctness for all J is not written out here. (iii) A control for the verification of Theorem 5.1, which assumed that the neighbouring residue $q \equiv 4 - 4J$ is *not* a zero on some even-J row, failed: $\eta_J(L(2J^2 - 1, 4 - 4J)) = 0$ on all eight even $J \in \{12, 14, 16, 20, 26, 30, 32, 40\}$ that were checked (all with composite p). In S-unit terms $4 - 4J \equiv 2\sqrt{2}\,\delta^{-1}$, the orbit $(1, -3)$; its behaviour is covered only by the period law of Section 7, and it is recorded as an observation, not a result.

6 The Pell ladder

Theorem 5.1 concerns the orbit of δ^2; the third family of [1, Thm. 7.1] is the orbit of δ (Lemma 4.1). Both are rungs of one ladder. Write

$$\begin{gathered} \delta^n = (1+\sqrt{2})^n = a_n + P_n\sqrt{2}, \\ (a_n)_{n\ge 0} = 1, 1, 3, 7, 17, 41, 99, \ldots, \qquad (P_n)_{n\ge 0} = 0, 1, 2, 5, 12, 29, 70, \ldots, \end{gathered} \tag{13}$$

so that $a_{n+1} = 2a_n + a_{n-1}$, $P_{n+1} = 2P_n + P_{n-1}$ and $a_n = P_n + P_{n-1}$, and put $\nu_n = (-1)^n$, the norm of δ^n. For $J \ge 2$ and $p = 2J^2 - 1$ let

$$q_n := a_n + 2P_n J, \tag{14}$$

the reduction of δ^n by (5), taken as a positive integer. Thus $q_1 = 2J + 1$ (family (b), the third family of [1, Thm. 7.1]) and $q_2 = 4J + 3 \equiv (3 - 4J)^{-1}$ (Theorem 5.1). Finally $M_n := 2P_n^2$.

Theorem 6.1 (The Pell ladder)**.** *Let $J \ge 2$, $p = 2J^2 - 1$ and q_n as in (14).*

(i) If $n \ge 1$ and $P_n \mid J$, then

$$\eta_J(L(p, q_n)) = \begin{cases} 0, & n \text{ odd}, \\ \frac{2}{3}\,P_{n-2}\,J, & n \text{ even}. \end{cases}$$

(ii) If $n \ge 2$ and $J \equiv 1 \pmod{P_n}$, then

$$\eta_J(L(p, q_n)) = \gamma_n\,\frac{J-1}{P_n}, \qquad \gamma_n = \begin{cases} 2P_{n-1}^2, & n \text{ even}, \\ \frac{4}{3}\,P_{n-1}a_{n-1}, & n \text{ odd}. \end{cases}$$

The same values hold on the orbit $\pm q_n^{\pm 1}$, with the sign of q (2).

Corollary 6.2. *(a) On the multiples of P_n, the orbit of δ^n is a zero of η_J iff n is odd or $n = 2$.*

(b) *For positive n, the orbit of δ^n vanishes at every rank $J \geq 2$, or even at all sufficiently large ranks, iff $n = 1$ (the third family of [1, Thm. 7.1]); for $n \geq 2$ part (ii) gives $\eta_J > 0$ on the infinite class $J \equiv 1 \pmod{P_n}$.*

(c) *$n = 2$ recovers Theorem 5.1 on both parities: 0 for even J by (i), and $\gamma_2(J-1)/2 = J-1$ for odd J by (ii).*

Proof. (a) For odd n the value in (i) is 0; for even n it is 0 iff $P_{n-2} = 0$, i.e. $n = 2$, since $P_{n-2} \geq 2$ for even $n \geq 4$. (b) For $n = 1$, $P_1 = 1$ divides every J and (i) gives 0; for $n \geq 2$, $\gamma_n > 0$. (c) $P_2 = 2$ and $\gamma_2 = 2P_1^2 = 2$. □

Lemma 6.3 (The Dedekind term at every rank)**.** *For $n \geq 1$, $J \geq 1$, $p = 2J^2 - 1$ and q_n as in (14),*

$$s(q_n, p) = \mathcal{T}(q_n, p) - \nu_n \, \mathcal{T}(M_n, q_n) + \nu_n \, s(q_n \bmod M_n, \, M_n), \tag{15}$$

and $q_n \bmod M_n = a_n + 2P_n \, (J \bmod P_n)$ depends on J only modulo P_n.

Pell facts. Fix $n \geq 1$ and write $a = a_n$, $b = P_n$, $\nu = \nu_n$, $M = M_n = 2b^2$. From $\delta^{n\pm1} = \delta^n \delta^{\pm1}$, $\delta^{-2} = 3 - 2\sqrt{2}$ and the norm:

(F1) $a^2 - M = \nu$.

(F2) $a_{n-1} = 2b - a \geq 1$, $P_{n-1} = a - b \geq 0$, $a_{n+1} = a + 2b$, $P_{n-2} = 3b - 2a$ for $n \geq 2$, and $P_{n-3} = 5a - 7b$ for $n \geq 3$.

(F3) $a_{n+1} a_{n-1} = M - \nu$.

(F4) $b \leq a < 2b$; a is odd and $\gcd(a, 2b) = 1$; and $2a \leq 3b$ for every $n \geq 1$, with equality only at $n = 2$ (for $n \geq 2$ this is $P_{n-2} \geq 0$; at $n = 1$ it follows from $a = b = 1$).

(F5) With $t := 2bJ - a$,

$$M \, p = q_n \, t + \nu, \tag{16}$$

because $M(2J^2 - 1) = (2bJ)^2 - a^2 + (a^2 - 2b^2)$. Hence $q_n \cdot (-\nu t) \equiv 1 \pmod{p}$, so $q_n^* \equiv -\nu t$ and $\rho_p(\alpha q_n^*) = -\nu \, \rho_p(\alpha t)$ for every integer α.

(F2) and (F3) are read off from $(a + b\sqrt{2})(\sqrt{2} \mp 1)$, $(a + b\sqrt{2})(3 - 2\sqrt{2})$ and $(a + 2b)(2b - a) = 4b^2 - a^2$; (F4) follows from (F1) and (F2) and the recurrence (a_n is odd).

Proof of Lemma 6.3. Write $q = q_n$. By (16), $Mp \equiv \nu \pmod{q}$, so $p \equiv \nu M^{-1} \pmod{q}$. Moreover $\gcd(M, q) = 1$: $q = a + 2bJ$ is odd, and modulo each prime factor of b one has $q \equiv a \not\equiv 0$ by (F4). Since $s(h^{-1}, k) = s(h, k)$ and $s(-h, k) = -s(h, k)$, this gives $s(p, q) = \nu \, s(M, q)$. Two applications of (8), $s(q, p) = \mathcal{T}(q, p) - s(p, q)$ and $s(M, q) = \mathcal{T}(M, q) - s(q, M)$, give (15), as $s(q, M)$ depends only on $q \bmod M$. Finally $2bJ \equiv 2b(J \bmod b) \pmod{2b^2}$ and $a + 2b(b-1) < 2b^2$ by (F4). □

What Lemma 6.3 does and does not give. With $r := a + 2b \, (J \bmod b)$ the identity (15) simplifies exactly to

$$s(q_n, p) = \frac{a}{6p} - \frac{a}{6M} + s(r, M) \quad (n \text{ even}), \qquad s(q_n, p) = \frac{bJ}{3p} + \frac{J}{6b} - \frac{1}{2} - s(r, M) \quad (n \text{ odd}). \tag{17}$$

Only the lookup $s(r, M)$ depends on J through $J \bmod P_n$; the rational terms do not vanish, and they are not eventually quasi-polynomial in J (for $n = 1$ there is a single residue class, yet $s(5, 7) = -\frac{1}{14}$

at $J=2$ and $s(7,17)=\frac{1}{17}$ at $J=3$). Lemma 6.3 is thus an exact reduction of the Dedekind term to a rational function of J plus a finite periodic lookup; it is not a proof of a "Dedekind-term part" of Conjecture 7.1 below. The cancellation of the rational remainders happens only in the complete η_J: on the two slices of Theorem 6.1 the residue term R_J absorbs them, and in the sectors of Section 8 the cancellation is proved through the integrality of η_J (Theorem 8.4).

Proof of Theorem 6.1. We evaluate the four terms of (7) at $q=q_n$, using (16) to control every residue; α denotes the summation index of R_J and (α,β) that of Δ_J.

Slice (i): $J=bk$, $k\geq 1$, $J\geq 2$. Here $q=Mk+a$, $t=Mk-a$ and $p=Mk^2-1$.

Step 1 (the Dedekind sum). $q \bmod M = a$, and by (F1) $M\equiv -\nu \pmod a$, so $s(a,M)=\mathcal{T}(a,M)+\nu\, s(1,a)$ with $s(1,a)=\frac{(a-1)(a-2)}{12a}$. Lemma 6.3 then simplifies to

$$s(q,p)=\frac{Mk+(k-3)p}{6p}\quad (n \text{ odd}),\qquad s(q,p)=\frac{a}{6p}+\frac{a-3}{6}\quad (n \text{ even}). \tag{18}$$

At $n=2$ this is $\frac{1}{2p}$, the even-J value in (10).

Step 2 (residues). For $1\leq\alpha\leq J-1$ write $\alpha=ck+r$ with $0\leq c\leq b-1$ and $0\leq r\leq k-1$. Then

$$\alpha q = cp+c(ak+1)+rq,\qquad \alpha t = cp-c(ak-1)+rt.$$

The residue $\langle\alpha q\rangle_p = c(ak+1)+rq$ never wraps: its largest value is $p-b\,a_{n-1}k+(b-a)<p$. The residue $\langle\alpha t\rangle_p$ equals $rt-c(ak-1)\in(0,p)$ for $r\geq 1$, and $p-c(ak-1)$ (one wrap) for $r=0$, $c\geq 1$. Hence

$$\rho_p(\alpha q)=p-2c(ak+1)-2rq,\qquad \rho_p(\alpha t)=p-2rt+2c(ak-1)-2p\,\mathbf{1}_{r=0},$$

and summing $(J-\alpha)[\rho_p(\alpha q)-\nu\rho_p(\alpha t)]$ over α (Faulhaber sums in c and r) gives

$$R_J=4J\,s(q,p)\quad (n \text{ odd}),\qquad R_J=4J\,s(q,p)+\tfrac{2}{3}P_{n-2}J\quad (n \text{ even}). \tag{19}$$

For odd n this is $R_J=4Js$, the cancellation pattern of the first three families of [1, Thm. 7.1]; for even n it reads $R_J=2b(b-1)k-\frac{4a}{3p}(J^3-J)$.

Step 3 (the triangular sum is empty). A term $\alpha q\equiv\beta$ with $1\leq\beta\leq J-1-\alpha$ would need $\langle\alpha q\rangle_p\leq J-1$; but $\langle\alpha q\rangle_p\geq q>J$ when $r\geq 1$ and $\langle\alpha q\rangle_p\geq ak+1>bk=J$ when $r=0$. A term $\alpha q\equiv-\beta$ would need $p-\langle\alpha q\rangle_p\leq J-1$; but $p-\langle\alpha q\rangle_p\geq b\,a_{n-1}k+(a-b)\geq J$. So $\Delta_J=0$.

Step 4 (the low-mode term vanishes). In the coordinates u,v of the proof of Theorem 5.1, the invariant low-mode weights are the lattice points with $|u|,|v|\leq J-1$ and $v+qu\equiv 0 \pmod p$. For $1\leq u\leq J-1$ such a point needs $\langle uq\rangle_p\leq J-1$ or $\langle uq\rangle_p\geq p-J+1$, both excluded by Step 3; negative u follow by symmetry, and $u=0$ forces $v=0$. Only $(0,0)$ remains, and it cancels in L_J as before. So $L_J=0$.

Assembly. $\eta_J=-4Js+R_J$, which is 0 for odd n and $\frac{2}{3}P_{n-2}J$ for even n; this is (i).

Slice (ii): $n\geq 2$, $J=bk+1$, $k\geq 1$. Write $a_-=a_{n-1}$ and $a_+=a_{n+1}$, so that $M=a_+a_-+\nu$ by (F3). Then $q=Mk+a_+$, $t=Mk+a_-$, $p=Mk^2+4bk+1$, and with

$$g:=a_-k+1,\qquad h:=a_+k+1,\qquad p=kq+g=kt+h.$$

Step 1 (the Dedekind sum). $q \bmod M=a_+$ (as $a_+<M$ for $b\geq 2$), and $M\equiv\nu \pmod{a_+}$ by (F3), so $s(a_+,M)=\mathcal{T}(a_+,M)-\nu\,s(1,a_+)$. Lemma 6.3 then simplifies to

$$s(q,p)=-\frac{a(J^2-1)}{3p}\quad (n \text{ even}),\qquad s(q,p)=\frac{J-1}{6b}-\frac{b(J-1)(2J+1)}{3p}\quad (n \text{ odd}). \tag{20}$$

At $n = 2$ this is $\frac{1}{2p} - \frac{1}{2}$, the odd-$J$ value in (10).

Step 2 (residues). For $1 \le \alpha \le J-1 = bk$ write $\alpha = ck + r$ with $0 \le c \le b$ and $0 \le r \le k-1$, where $c = b$ occurs only with $r = 0$. Then $\alpha q = cp + rq - cg$ and $\alpha t = cp + rt - ch$. The residue $\langle \alpha q \rangle_p$ equals $rq - cg \in (0, p)$ for $r \ge 1$ and $p - cg$ for $r = 0$. For $\langle \alpha t \rangle_p$ put $c^* := a_- + \mathbf{1}_{n \text{ even}}$; then $\langle \alpha t \rangle_p = rt - ch$ for $r \ge 2$ and for $r = 1$ with $c < c^*$; $= p + rt - ch$ for $r = 1$ with $c \ge c^*$; and $= p - ch$ for $r = 0$. The sign at $r = 1$ comes from

$$t - ch = (a_+(a_- - c) + \nu)k + (a_- - c).$$

For $k = 1$ there is no row $r = 1$; instead $\langle \alpha t \rangle_p = 2p - ch$ exactly for $c^* + 1 \le c \le b$, because $c^* h \le p < (c^* + 1)h$ at $k = 1$. That correction equals the $r = 1$ correction of the formula for $k \ge 2$, so one closed form covers every $k \ge 1$. Summing as in slice (i),

$$R_J = 4J\, s(q,p) + \kappa_n k, \qquad \kappa_n = \begin{cases} 2P_{n-1}(P_{n-1} - 1), & n \text{ even}, \\ 2P_{n-1}(P_{n-1} - 1) - \frac{2}{3}P_{n-1}P_{n-3}, & n \text{ odd } (n \ge 3). \end{cases} \tag{21}$$

Step 3 (the triangular sum). A term $\alpha q \equiv +\beta$ is impossible: $\langle \alpha q \rangle_p > J$ when $r \ge 1$ and $\langle \alpha q \rangle_p \ge p - bg > J$ when $r = 0$. For $\alpha q \equiv -\beta$ with $r \ge 1$, $p - \langle \alpha q \rangle_p \ge g + q > J$. For $r = 0$, $\alpha = ck$ and $\beta = cg$, and $\alpha + \beta \le J - 1$ reads $c(1 + a_-)k + c \le bk$; this holds only for $c = 1$, since $2(1 + a_-) > b$ by (F4), and it needs $P_{n-1} = b - a_- \ge 2$. So there is exactly one pair, $(\alpha, \beta) = (k,\ a_- k + 1)$, of type $\alpha q + \beta \equiv 0$, with weight $J - \alpha - \beta = (P_{n-1} - 1)k$, and

$$\Delta_J = -(P_{n-1} - 1)\, k, \tag{22}$$

which is 0 for $n = 2$.

Step 4 (the low-mode term). By the same bounds the only nonzero lattice points with $|u|, |v| \le J - 1$ and $v + qu \equiv 0$ are $\pm(k,\ a_- k + 1)$; the point with $c = 2$ would need $2(a_- k + 1) \le bk$, false by (F4). In weights, $(w_L, w_R) = \big((1 + a_-)k + 1,\ (a_- - 1)k + 1\big)$, both odd since a_- is odd. This point lies in $W_{J-m-1} \times W_{J+m-1}$ for $m \equiv J \pmod 2$ with $1 \le m \le (P_{n-1} - 1)k - 1$, and in $W_{J+m-1} \times W_{J-m-1}$ for $m \equiv J \pmod 2$ with $1 \le m \le (P_{n-1} + 1)k - 1$. The two ranges differ by exactly k values of m (an interval of $2k$ consecutive integers, or $[1, 2k-1]$ for $n = 2$ with J odd). Counting both points $\pm$ with the sign of L_J,

$$L_J = -2k. \tag{23}$$

Assembly. By (7) and (21)–(23),

$$\eta_J = (\kappa_n + 2(P_{n-1} - 1) + 2)k = \gamma_n k,$$

using $P_{n-3} = P_{n-1} - 2P_{n-2}$ and $a_{n-1} = P_{n-1} + P_{n-2}$ for odd n; and $k = (J-1)/P_n$. This is (ii). The orbit statement is (2). □

Remark 6.4 (Certificate for the finite sums)**.** The closed forms (18), (19), (20) and (21) are identities between rational functions of (a, b, k) modulo the relation $a^2 = 2b^2 + \nu$; they follow from the residue structure of Steps 1–2 by Faulhaber sums, which are not written out above. They were checked by two independent routes: with sympy over exact rationals (polynomial remainder modulo $a^2 - 2b^2 - \nu$, no numerical simplification) and with the local Wolfram Engine 14.3 (`PolynomialRemainder`), four identity sets on each route, all true. Every step was also checked numerically on every row of Appendix A. The proof of Theorem 6.1 is certificate-assisted at exactly this point; the residue analysis, the emptiness and one-pair statements of Step 3, and the lattice counts of Step 4 are as written. (The certificates of Section 8 also have both routes; see Remark 8.11.)

Remark 6.5 (Verification, and the size of the rungs)**.** Theorem 6.1 was verified term by term on 212 rows, 81 of which were never computed before the statement was fixed, up to $n = 16$; the largest are $J = 2P_{16} = 941{,}664$, $p = 1{,}773{,}462{,}177{,}791$ in slice (i), with $\eta_J = 50{,}713{,}000{,}832 = \frac{2}{3}P_{14}J$, and $J = 941{,}665$ in slice (ii), with $\Delta_J = -390{,}048 = -(P_{15}-1)\cdot 2$, $L_J = -4$ and $\eta_J = 152{,}139{,}002{,}500 = 2\gamma_{16}$. Lemma 6.3 reproduces the value of Corollary 2.8 on every $n = 2$ row. A mutated modulus, "$\eta_J = 0$ on $J \equiv 0 \pmod{a_n}$", fails at every n tested, for instance $\eta_J = 284, -1144, 1152$ at $J = 707, 714, 721$ ($n = 3$).

7 The period law

Theorem 6.1 says that on the orbit of δ^n the value of η_J is governed by J modulo P_n, the coefficient of $\sqrt{2}$. Exploratory calculations (Observation 7.3 below) suggested that this is the general shape. For an S-unit orbit choose a representative $\xi = x + y\sqrt{2} \in \mathbb{Z}[\sqrt{2}]$ with $y \neq 0$; its reduction is $\xi(J) = x + 2yJ$, and by (2) the choice of representative changes η_J at most by sign.

Conjecture 7.1 (Period law)**.** *For every S-unit orbit with representative $\xi = x + y\sqrt{2}$, $y \neq 0$, the function $J \mapsto \eta_J(L(2J^2 - 1,\, x + 2yJ))$ is, for $J \geq J_0(\xi)$, a quasi-polynomial in J whose period divides $|y|$.*

Status. Conjecture 7.1 is proved for every orbit of norm ± 1 (the silver powers, $m = 0$) and of norm ± 2 ($|m| = 1$) to which it applies, that is with $y \neq 0$; this excludes only the constant orbit $q = \pm 1$, where η_J is cubic in J (Lemma 8.8): Theorem 8.4 in Section 8, with Lemma 6.3 and Theorem 6.1 as its first ingredients. For $|m| \geq 2$, for example the fourth family $2\delta = (1, 2)$ and the orbit $(1, -3)$ of $q \equiv 4 - 4J$, it remains a conjecture, supported by the held-out test below.

Held-out numerical test. Let $\mathcal{W}$ be the set of 64 orbits (n, m) of $\delta^n\sqrt{2}^{\,m}$ with $0 \leq n \leq 5$, $|m| \leq 9$ ($m \geq 0$ when $n = 0$), $y \neq 0$ and $|y| \leq 41$; here $|y| = 2^{\lfloor |m|/2 \rfloor} P_n$ for even m and $2^{\lfloor |m|/2 \rfloor} a_n$ for odd m. On $J \in [150, 400]$ every orbit of $\mathcal{W}$ was fitted by one polynomial per residue class of J modulo $|y|$ (through the first four points of the class, verified on the rest). These fits were then used to predict η_J for every orbit of $\mathcal{W}$ and every $J \in [1401, 1600]$, 12,800 values, before those rows were computed; all 12,800 predictions were exact (only the summary counts of this test were retained, not the individual values). This is finite evidence for Conjecture 7.1; it does not locate the onset $J_0(\xi)$, and nothing below uses it as a proof.

Corollary 7.2 (Window)**.** *(a)* (Every rank; proved.) *Among the 64 orbits of $\mathcal{W}$, exactly four vanish at every rank $J \geq 2$: the orbits $(0, 1)$, $(1, -1)$, $(1, 0)$, $(1, 2)$ of Lemma 4.1, i.e. the four families of [1, Thm. 7.1].*

(b) (All large ranks.) *For the orbits of $\mathcal{W}$ with $|m| \leq 1$ (norm ± 1 or ± 2), none but $(0, 1)$, $(1, -1)$, $(1, 0)$ vanishes at all sufficiently large ranks; this is proved (Corollary 8.9 below). For the orbits with $|m| \geq 2$, which include the fourth family $(1, 2)$, the corresponding statement is observed on $J \in [150, 400] \cup [1401, 1600]$ and follows from Conjecture 7.1 only together with the assumption that the onset $J_0(\xi)$ of each such orbit lies below 150.*

Proof. (a) The four orbits vanish at every $J \geq 2$ by [1, Thm. 7.1]. For each of the other 60 orbits a single exact nonzero value refutes vanishing at every rank. The following ledger gives one such witness (J, q, η_J) per orbit (n, m), with $p = 2J^2 - 1$ and q the reduction of $\delta^n\sqrt{2}^{\,m}$ modulo p; every witness has $J \leq 6$, and each value was cross-checked against the independent evaluator of Appendix A.

(n,m)	J	q	η_J	(n,m)	J	q	η_J	(n,m)	J	q	η_J
$(0,3)$	2	1	-2	$(2,-8)$	3	2	-2	$(3,-3)$	2	6	2
$(0,5)$	4	1	-28	$(2,-7)$	2	1	-2	$(3,-2)$	5	4	-2
$(0,7)$	4	2	-8	$(2,-6)$	4	14	2	$(3,-1)$	3	9	-2
$(0,9)$	2	1	-2	$(2,-5)$	6	5	-2	$(3,0)$	2	6	2
$(1,-9)$	4	10	2	$(2,-4)$	2	1	-2	$(3,1)$	3	1	-10
$(1,-8)$	2	6	2	$(2,-3)$	5	33	-8	$(3,2)$	4	1	-28
$(1,-7)$	3	9	-2	$(2,-2)$	3	16	10	$(3,3)$	2	6	2
$(1,-6)$	5	32	4	$(2,-1)$	2	1	-2	$(3,4)$	4	2	-8
$(1,-5)$	2	6	2	$(2,0)$	3	15	2	$(3,5)$	4	16	-8
$(1,-4)$	4	10	2	$(2,1)$	4	28	2	$(3,6)$	2	6	2
$(1,-3)$	3	2	-2	$(2,2)$	2	1	-2	$(4,-3)$	5	24	20
$(1,-2)$	2	6	2	$(2,3)$	6	9	4	$(4,-2)$	2	1	-2
$(1,1)$	2	6	2	$(2,4)$	3	9	-2	$(4,-1)$	5	48	60
$(1,3)$	3	16	10	$(2,5)$	2	1	-2	$(4,0)$	4	20	2
$(1,4)$	2	6	2	$(2,6)$	3	1	-10	$(4,1)$	2	1	-2
$(1,5)$	3	15	2	$(2,7)$	6	36	-40	$(4,2)$	3	8	2
$(1,6)$	4	10	2	$(2,8)$	2	1	-2	$(4,3)$	4	10	2
$(1,7)$	2	6	2	$(3,-6)$	2	6	2	$(5,-1)$	2	6	2
$(1,8)$	4	20	2	$(3,-5)$	3	15	2	$(5,0)$	5	37	-2
$(1,9)$	3	9	-2	$(3,-4)$	5	2	-20	$(5,1)$	3	15	2

(b) The first statement is Corollary 8.9. For the second, an orbit that vanishes at all sufficiently large ranks has, under Conjecture 7.1, identically zero class polynomials from its onset on; if the onset lies below 150 these are the fitted polynomials, which are not all zero for any of the other 60 orbits. □

The every-rank statement (a) rests on exact witnesses and the four all-rank proofs of [1]; it uses neither Conjecture 7.1 nor the held-out test.

Observation 7.3 (The zero set at $p = 2J^2 - 1$; exploratory data)**.** (a) On $J \in [150, 400]$ each orbit of $\mathcal{W}$ is a quasi-polynomial in J of degree at most 2 on each residue class modulo $|y|$. The period-2 zero families in $\mathcal{W}$ are $(2,0)$ (Theorem 5.1: even J), $(1,-3)$ (Remark 5.2(iii): even J), $(1,-2)$ (odd J) and $(0,3)$ (odd J). Larger periods give scattered zero classes: δ^5 vanishes for $J \equiv 0, \pm 10 \pmod{29}$ and δ^7 for $J \equiv 0, \pm 24, \pm 46, \pm 75 \pmod{169}$.

(b) The zero set is not confined to S-units. At the ranks $J = 20, 30, 40$ ($p = 799, 1799, 3199$) the numbers of units q with $\eta_J(L(p,q)) = 0$ are 120, 180 and 268, among the 734, 1,534 and 2,734 units $q \not\equiv \pm 1$: about $6J$ zeros per rank. About 41% of the units have no nonzero lattice point with $|u|, |v| \le J - 1$; between 95% and 100% of the zeros at these ranks are of that kind, and for them η_J equals $-4Js + R_J$ alone.

(c) What these counts do and do not show. The zero set contains infinitely many congruence families (Theorem 6.1); no S-unit orbit of norm ± 1 or ± 2 other than the first three families of [1, Thm. 7.1] vanishes at every rank (Corollary 8.9); and in $\mathcal{W}$ exactly the four families vanish at every rank (Corollary 7.2(a)). Whether the full zero set at $p = 2J^2 - 1$ admits a finite parametrized description is open: an infinite indexed ladder can itself have one finite defining formula, and the counts at three ranks do not decide this.

Remark 7.4 (A refuted "zero-term principle")**.** One might guess that on a zero class each of $-4Js + R_J$, Δ_J and L_J vanishes separately, as in slice (i) of Theorem 6.1. This is false: in 52 of 4,383 zero-class rows examined, all in the orbits $(4, \pm 2)$, the term $-4Js + R_J$ cancels against L_J

instead. Zeros of η_J can arise from cancellation between the Dedekind–residue part and the lattice part.

Remark 7.5 (Narrowing of Observation 4.4)**.** Observation 4.4(b) describes statistics in the coordinate ρ at primes taken one silver rank each. It does not test congruence classes of ranks along $p = 2J^2 - 1$: in the coordinate q the silver-power orbits carry structured zeros on such classes (Theorem 6.1). Thus those finite coordinate statistics do not justify describing these zeros as accidental. No structural claim is made about the remaining sporadic zeros, and Observation 4.4 remains a description of its stated ranges.

8 The period law in the unit and norm-two sectors

Conjecture 7.1 is a theorem for every nonconstant S-unit orbit of norm ± 1 or ± 2, that is every such orbit with $y \neq 0$. Concretely, with $a = a_n$, $b = P_n$, $\nu = \nu_n$ as in Section 6, the orbits of Theorem 8.4 are

- the *unit sector* S_0: $q = a_n + 2P_nJ = q_n$, the reduction of δ^n, $n \geq 1$, of period P_n;
- the *norm-two sector* S_1: $q = a_nJ + \sigma P_n$ with $\sigma = \pm 1$, $n \geq 2$, of period a_n; here $\sigma = +1$ is the reduction of $\delta^n/\sqrt{2}$, the orbit $(n, -1)$, and $\sigma = -1$ is, up to the sign ν, the reduction of $\delta^{-n}/\sqrt{2}$, the orbit $(n, 1)$ of $\delta^n\sqrt{2}$;

and the small nonconstant orbits $q = J$ and $q = J \pm 1$ ($n = 0, 1$ in S_1, period 1) are covered by Theorem 2.7, [1, Thm. 7.1] and Lemma 8.7, where η_J is 0, 0 and -2. The constant orbit $q = \pm 1$ ($n = 0$ in S_0, $y = 0$) lies outside Conjecture 7.1; there $\eta_J(L(2J^2 - 1, 1)) = -J(J-1)(2J-1)/3$ is cubic in J (Lemma 8.8), and no statement below about linear class polynomials applies to it. An orbit (n, m) has norm ± 1 iff $m = 0$ and contains an element of norm ± 2 iff $|m| = 1$; the fourth family $2\delta = (1, 2)$ has norm -4 and lies outside this section. The proof has four ingredients: Lemma 6.3 and its norm-two analogue make the Dedekind term a rational function of k on each residue class; a breakpoint lemma does the same for the residue term; Theorem 2.4 bounds the denominators of η_J, which forces a rational function to be a polynomial; and a box form of the two lattice terms of (3) shows that they are eventually linear in k. Throughout, $\Lambda_q = \{(u, v) \in \mathbb{Z}^2 : \ v + qu \equiv 0 \pmod{p}\}$, $\|(u, v)\| = \max(|u|, |v|)$, and $\langle x \rangle$ is the least non-negative residue modulo p.

Lemma 8.1 (Box form of the lattice terms)**.** *For every $p \geq 2$, every unit q modulo p and every $J \geq 1$,*

$$-2\Delta_J - L_J = \sum_{\substack{(u,v)\in\Lambda_q\setminus\{0\} \\ \|(u,v)\|\leq J-1}} \operatorname{sgn}(uv)\,(J - \|(u, v)\|) =: \mathcal{B}_J(q). \tag{24}$$

Hence $\eta_J(L(p, q)) = [-4J\, s(q, p) + R_J] + \mathcal{B}_J(q)$. This is a rewriting of the two lattice terms of (3), not a new invariant.

Proof. The low-mode term. By the substitution of the proof of Theorem 5.1, $w_L = u + v$, $w_R = v - u$, the pairs counted by $I_{J-m,J+m}$ are the lattice points $(u, v) \in \Lambda_q$ with $|u + v| \leq J - m - 1$, $|v - u| \leq J + m - 1$ and $u + v \equiv J - m - 1 \pmod 2$, and those counted by $I_{J+m,J-m}$ are the same with the two bounds exchanged; every such point has $\|(u, v)\| \leq J - 1$. Fix a nonzero $(u, v) \in \Lambda_q$ with $M := \|(u, v)\| \leq J - 1$, put $m_0 := \min(|u|, |v|)$, $r = u + v$, $t = v - u$ (so $r \equiv t \pmod 2$), and count the $m \in [1, J - 1]$ for which the point is counted. If $|r| \leq J$ and $|t| \leq J$, the two counts are $\lfloor (J - |r|)/2 \rfloor$ and $\lfloor (J - |t|)/2 \rfloor$, whose difference is $(|t| - |r|)/2 = -\operatorname{sgn}(uv)\, m_0$. Otherwise exactly

one of the two counts is nonzero and equals $J-M$, the count with the sign $-\operatorname{sgn}(uv)$. Hence the point contributes $-\operatorname{sgn}(uv)\min(m_0,\,J-M)$ to L_J; points with $uv=0$ contribute nothing.

The triangular sum. A pair (a,b) with $aq\equiv b$ is the point $(a,-b)\in\Lambda_q$ and a pair with $aq\equiv -b$ is the point (a,b), with weight $J-|u|-|v|$ when $|u|+|v|\le J-1$; so, using the symmetry $(u,v)\mapsto(-u,-v)$ of Λ_q and of $\operatorname{sgn}(uv)$, $-2\Delta_J=\sum\operatorname{sgn}(uv)\,(J-|u|-|v|)_+$ over the nonzero points of Λ_q.

Sum. If $m_0+M\le J-1$ the two contributions add to $(J-M-m_0)+m_0=J-M$; otherwise $(J-|u|-|v|)_+=0$ and $\min(m_0,J-M)=J-M$. □

Lemma 8.2 (Norm-two Dedekind reduction)**.** *Let $n\ge 1$, $J\ge 2$, $p=2J^2-1$, $\sigma=\pm1$, $q=aJ+\sigma b$ and $t=aJ-\sigma b$. Then $a^2p=2qt-\nu$, hence $q^*\equiv 2\nu t \pmod p$, and*

$$s(q,p)=\mathcal{T}(q,p)+\nu\,\mathcal{T}(a^2,q)-\nu\,s(q \bmod a^2,\,a^2), \tag{25}$$

where $q \bmod a^2=(aJ+\sigma b) \bmod a^2$ depends only on $J \bmod a$.

Proof. $2qt=2(a^2J^2-b^2)=a^2(p+1)-2b^2=a^2p+\nu$ by (F1). So $2qt\equiv\nu$ modulo p, which gives q^*, and $a^2p\equiv-\nu \pmod q$, so $p\equiv-\nu a^{-2} \pmod q$; here $\gcd(a,q)=\gcd(a,b)=1$. Hence $s(p,q)=-\nu\,s(a^2,q)$, and two applications of (8) give (25). □

Lemma 8.3 (Breakpoints)**.** *Let $k\ge 1$.*

(a) (S_0.) *Let $J=bk+j$ with $0\le j<b$, $q=q_n$, $t=2bJ-a$, $d_j=2bj-a$ and $q_j=a+2bj$. For $1\le c\le b$, $\lceil cp/q\rceil=ck+e_c$ with $e_c=\lceil c\,d_j/(2b^2)\rceil$, or $c\,d_j/(2b^2)+\mathbf{1}_{\nu=1}$ when that quotient is an integer; and $\lceil cp/t\rceil=ck+e'_c$, the same with q_j in place of d_j.*

(b) (S_1.) *Let $J=ak+j$ with $0\le j<a$, $q=aJ+\sigma b$, $t=aJ-\sigma b$. Then $\lceil cp/q\rceil=2ck+\lceil 2c(aj-\sigma b)/a^2\rceil$ for $1\le c\le(a+1)/2$ and $\lceil cp/(2t)\rceil=ck+\lceil c(aj+\sigma b)/a^2\rceil$ for $1\le c\le a$, both ceilings being of non-integers.*

(c) In both sectors $q^*\equiv-\nu t$ (S_0) or $2\nu t$ (S_1). The number of wraps $\lfloor\alpha x/p\rfloor$ of a residue $\langle\alpha x\rangle$, $1\le\alpha\le J-1$, equals the number of the breakpoints $\lceil cp/x\rceil$ that are $\le\alpha$. The breakpoints entering the layer-cake sums below ($c\le b-1$ in S_0; $c\le(a-1)/2$ for q and $c\le a-1$ for $2t$ in S_1) are all $\le J+1$, and in S_1 all $\le J$, whenever $J\ge P$ (the period). The S_1 breakpoint $c=(a+1)/2$ of q is listed only to show that it is $\ge J$; it is inactive and can exceed $J+1$ (for $a=3$, $b=2$, $\sigma=-1$, $J=3$ it is $\lceil 34/7\rceil=5$).

Proof. (a) From $M_np=qt+\nu$ (16) and $t=M_nk+d_j$, $q=M_nk+q_j$: $cp/q=ck+c\,d_j/M_n+c\nu/(M_nq)$ and $cp/t=ck+c\,q_j/M_n+c\nu/(M_nt)$. The last terms have absolute value $<1/M_n$ because $c<q$ and $c<t$, so the ceiling is decided by $c\,d_j/M_n$ (resp. $c\,q_j/M_n$), or, when that is an integer, by the sign of ν. Also $\lceil bp/q\rceil=J$ and $\lceil bp/t\rceil=J+1$, because $0<a/(2b)<1$. (b) From $a^2p=2qt-\nu$ (Lemma 8.2) and $t=a^2k+aj-\sigma b$, $q=a^2k+aj+\sigma b$: $cp/q=2ck+2c(aj-\sigma b)/a^2-c\nu/(a^2q)$ and $cp/(2t)=ck+c(aj+\sigma b)/a^2-c\nu/(2a^2t)$. In the stated ranges a^2 divides neither $2c(aj-\sigma b)$ nor $c(aj+\sigma b)$, because every prime $\ell\mid a$ has $\ell\nmid b$, hence $\ell\nmid(aj\mp\sigma b)$; so the ceilings are of non-integers, and the corrections are below $1/a^2$ in absolute value. (c) The inverses are (16) and Lemma 8.2; the wrap count is the definition of the breakpoints. In S_1 the breakpoint $c=(a+1)/2$ of q is at least $(a+1)k+j-1>J-1$, and $\lceil ap/(2t)\rceil\ge J$; in S_0 the last breakpoints are J and $J+1$ by (a), and they bound every earlier active breakpoint by monotonicity in c. For the active S_1 breakpoints we use the following identities: with a odd,

$$Jq-\tfrac{a-1}{2}\,p=J^2+\sigma bJ+\tfrac{a-1}{2},\qquad J(2t)-(a-1)\,p=2J^2-2\sigma bJ+a-1,$$

both positive for $J \geq a > b$. Hence $\lceil \frac{a-1}{2}\, p/q \rceil \leq J$ and $\lceil (a-1)p/(2t) \rceil \leq J$, and by monotonicity in c every active breakpoint is $\leq J$. The layer-cake form is used below only for $k \geq 1$, that is for $J \geq P$, which in S_1 is $J \geq a$. □

The finite breakpoint checks, on every class $j < 40$ with $k \leq 5$ of the 24 families of Observation 8.10, are reported separately in Appendix A.

Theorem 8.4 (The period law in S_0 and S_1)**.** *Let the family be S_0 ($n \geq 1$) or S_1 ($n \geq 2$, $\sigma = \pm 1$), with period $P = P_n$ (S_0) or $P = a_n$ (S_1). For each residue j modulo P there is a polynomial $\mathcal{Q}_j$ of degree at most 1 and a threshold $k_j^* \geq 1$, computed from linear data of the class, such that*

$$\eta_J(L(p,q)) = \mathcal{Q}_j(k) \qquad \text{for every } J = Pk + j \text{ with } k \geq k_j^*. \tag{26}$$

A class is a zero class ($\mathcal{Q}_j \equiv 0$) iff $\mathcal{Q}_j$ vanishes at k_j^* and $k_j^* + 1$. The complete zero set of the orbit is the union of the zero classes (for $k \geq k_j^*$), the isolated zeros (integer roots $k \geq k_j^*$ of the nonzero $\mathcal{Q}_j$, at most one per class), and the finitely many zeros with $k < k_j^*$, which are checked directly. A nonzero class polynomial can vanish at an isolated rank, so the zero set need not be a union of whole residue classes beyond the thresholds.

Proof. Write $\eta_J = f(k) + \mathcal{B}_J$ with $f = -4Js + R_J$ (Lemma 8.1), and fix the class j.

(1) The bulk f is a rational function of k for every $k \geq 1$. The Dedekind term is given by Lemma 6.3 (S_0) or Lemma 8.2 (S_1); its only arithmetic input is a Dedekind sum of fixed modulus ($2b^2$ or a^2) at an argument fixed by j, and the rest is the rational function $\mathcal{T}$. The residue term is $\frac{2}{p}\sum_\alpha (J-\alpha)\rho_p(\alpha q)$ plus the inverse part, which is $-\nu\frac{2}{p}\sum_\alpha (J-\alpha)\rho_p(\alpha t)$ in S_0 and $\nu\frac{2}{p}\sum_\alpha (J-\alpha)\rho_p(2\alpha t)$ in S_1 (Lemma 8.3(c)). Here $\rho_p(\alpha x) = p\,(1+2c_x(\alpha)) - 2\alpha x$ with $c_x(\alpha)$ the number of wraps, and by Lemma 8.3 and the layer-cake identity $\sum_\alpha (J-\alpha)c_x(\alpha) = \sum_\beta (J-\beta)(J-\beta+1)/2$ over the active breakpoints β of Lemma 8.3(c) (all $\leq J+1$, so a breakpoint equal to J or $J+1$ contributes 0), $p\,R_J$ is a polynomial in k.

(2) $f(k) \in \frac{2}{3}\mathbb{Z}$ for every $k \geq 1$. By Theorem 2.4 at $p = 2J^2 - 1$, $\eta_J \in -4Jp\,s(q,p) + 2\mathbb{Z}$, and $6p\,s(q,p) \in \mathbb{Z}$ by Lemma 2.3, so $\eta_J \in \frac{2}{3}\mathbb{Z}$; and $\mathcal{B}_J \in 2\mathbb{Z}$, because its summand is invariant under $(u,v) \mapsto (-u,-v)$ and $(0,0)$ is excluded, so the points pair up. (An integer $\mathcal{B}_J$ would already give $f \in \frac{1}{3}\mathbb{Z}$, which also suffices below.)

(3) Rational to polynomial. A rational function $g \in \mathbb{Q}(k)$ with $g(k) \in \frac{1}{D}\mathbb{Z}$ for all large integers k is a polynomial: write $g = S + R/Q$ with $\deg R < \deg Q$; then $D'\,R(k)/Q(k) \in \mathbb{Z}$ for large k and tends to 0, so $R \equiv 0$. Growth: $s = O(J)$ by Lemma 6.3 or 8.2, so $f = O(k^2)$ and $\deg f \leq 2$.

(3′) The k^2 coefficient of f vanishes. Write $\rho_p(\alpha x) = p(1 + 2c_x(\alpha)) - 2\alpha x$ with $c_x(\alpha)$ the number of wraps. Then R_J is the sum of a constant part $2(1 \mp \nu)\sum_\alpha (J-\alpha)$ (upper sign S_0), the layer-cake part $4\sum_\alpha (J-\alpha)(c_q(\alpha) \mp \nu\, c_x(\alpha))$ with $x = t$ (S_0) or $x = 2t$ and the lower sign (S_1), and the part $-\frac{4}{p}\sum_\alpha (J-\alpha)\alpha\,(q - \nu t)$, resp. $(q + 2\nu t)$, with $\sum_\alpha (J-\alpha)\alpha = (J^3 - J)/6$. The breakpoints are linear in k with bounded offsets (Lemma 8.3), and the Dedekind lookups are bounded, so only leading slopes enter the k^2 coefficient $\lim f/k^2$. Part by part it is

	$-4J\cdot$(rational part of s)	constant	layer cake	α-part	sum
S_0, $\nu = +1$	0	0	0	0	0
S_0, $\nu = -1$	$-\frac{2b}{3}$	$2b^2$	$\frac{2}{3}(b-1)b(2b-1)$	$-\frac{4}{3}b^3$	0
S_1, $\nu = +1$	$-a$	$2a^2$	$a(a-1)^2$	$-a^3$	0
S_1, $\nu = -1$	$-\frac{a}{3}$	0	$-\frac{1}{3}(a-1)a(a+1)$	$\frac{a^3}{3}$	0

independently of σ, so $\deg f \leq 1$. (The values were derived by hand and certified symbolically by sympy and by the local Wolfram Engine 14.3, with identical part values; Remark 8.11.)

(4) The box $\mathcal{B}_J$ is linear in k for $k \geq k_j^*$. Within one wrap interval $\langle \alpha q \rangle$ increases by $q > J-1$, so a lattice point of the box with $u = \alpha \geq 1$ sits at an interval start ($\langle \alpha q \rangle \leq J-1$, $v = -\langle \alpha q \rangle$) or at an interval end ($p - \langle \alpha q \rangle \leq J-1$, $v = p - \langle \alpha q \rangle$); negative u follow by symmetry, and $u = 0$ gives $v = 0$. At the starts and at the regular ends ($\alpha = \beta - 1$ for a breakpoint $\beta \leq J$) both u and $|v|$ are linear in k: $\langle \alpha_c q \rangle = \alpha_c q - cp$ has no k^2 term, and $p - \langle (\beta-1)q \rangle = q - \langle \beta q \rangle$. The last element $u = J-1$ is a regular end, except in S_1 when it lies in the interval $c = (a-1)/2$ truncated at $J-1$; there $p - \langle (J-1)q \rangle = \frac{a+1}{2}p - (J-1)q = J^2 + (a - \sigma b)J + \sigma b - \frac{a+1}{2}$, which exceeds $J-1$ for $J \geq a$ because $a^2 > b + (a-1)/2$, so it is never in the box (if that breakpoint exceeds $J-1$, which happens only at $k = 1$, it equals J and $J-1$ is a regular end). Now $k_j^* \geq 1$ is the first $k \geq 1$ beyond every crossing of the linear conditions $|v| \leq J-1$, $u \leq J-1$ and $u \lneqq |v|$; each linear form is fixed by three exact values, and collinearity certifies that its k^2 term is absent. For $k \geq k_j^*$, $\mathcal{B}_J$ is a fixed signed sum of terms $J - \max(u, |v|)$, hence linear in k.

(5) $\eta_J = f + \mathcal{B}_J$ is a polynomial of degree at most 1 for $k \geq k_j^*$; two values determine it. □

The decision procedure. For a family and a class j, compute the threshold k_j^* from the linear data of step (4), evaluate η_J exactly at $k = k_j^*, k_j^* + 1, k_j^* + 2$, and read off $\mathcal{Q}_j$; the class is a zero class iff the values vanish (two suffice by the degree bound; the recorded computations used three). For the complete zero set, also solve each nonzero $\mathcal{Q}_j$ for integer roots $k \geq k_j^*$ and evaluate the finitely many ranks with $k < k_j^*$ directly. The zero classes for S_0, $n \leq 10$, and S_1, $n \leq 8$, are in Observation 8.10. Across these 24 families there is exactly one isolated zero beyond the thresholds: S_1, $n = 7$, $\sigma = +1$, class $j = 190$, where $\mathcal{Q}_{190}(k) = 2 - 2k$ vanishes at $k = 1$, $J = 429$ (and $\eta = -2, -4$ at $J = 668, 907$). There are also 363 zeros below the thresholds, mostly at the smallest rank of a class. The isolated zero and all 364 zeros were re-evaluated exactly; the complete sets are retained with the sources (`results/LC14B_ZERO_SETS.json` of [7]).

Corollary 8.5 (Eventual periodicity of the zero set)**.** *In each family of Theorem 8.4, with period P, put*

$$J_1 = \max_j \Big(P\,(k_j^* + 1) + j,\ \max_r \big(P\,(r+1) + j \big) \Big),$$

where r runs over the integer roots of the nonzero $\mathcal{Q}_j$ (the inner maximum is omitted when there are no such roots). For $J \geq J_1$, $\eta_J(L(2J^2 - 1, q)) = 0$ iff $J \bmod P$ is a zero class. Hence the zero set of the orbit is eventually periodic, and its natural density is (number of zero classes)/P.

Proof. For $J = Pk + j \geq J_1$ one has $k > k_j^*$ and k exceeds every root of $\mathcal{Q}_j$, so $\eta_J = \mathcal{Q}_j(k)$ vanishes iff $\mathcal{Q}_j \equiv 0$. □

The periodic onset need not equal a class threshold: the isolated zero $J = 429$ lies past its class threshold but below J_1. Regular class laws and one-off cancellations coexist. For example, the densities of the zero sets of $\delta^3, \ldots, \delta^9$ are 1/5, 2/12, 3/29, 10/70, 7/169, 18/408 and 3/985, read off from Observation 8.10.

Theorem 8.6 (The norm-two slice $J \equiv 0 \pmod{a_n}$)**.** *For every $n \geq 2$, $k \geq 1$ and $J = a_n k$, $p = 2J^2 - 1$,*

$$\eta_J(L(p,\, a_n J - P_n)) = \begin{cases} -P_{n-1}^2\, k, & n \text{ odd}, \\ \frac{1}{3} P_n P_{n-2}\, k, & n \text{ even}, \end{cases} \qquad \eta_J(L(p,\, a_n J + P_n)) = -\eta_J(L(p,\, a_n J - P_n)).$$

It is zero only for $n = 2$.

Proof. This is the class $j = 0$ of S_1. By Lemma 8.2, since $q \bmod a^2 \equiv \sigma b$ and $a^2 \equiv \nu \pmod b$ by (F1), $s(q,p) = \mathcal{T}(q,p) + \nu\mathcal{T}(a^2,q) - \nu\sigma\mathcal{T}(b,a^2) + \sigma s(1,b)$. Lemma 8.3(b) gives the breakpoints $2ck + \mathbf{1}_{\sigma=-1}$ for q, $1 \le c \le (a-1)/2$, and $ck + \mathbf{1}_{\sigma=1}$ for $2t$, $1 \le c \le a-1$. The box is empty for every $k \ge 1$: at the interval starts $\langle \alpha_c q \rangle = k(2c\sigma b + \mathbf{1}_{\sigma=-1}a^2) + (\mathbf{1}_{\sigma=-1}\sigma b + c)$ exceeds $J-1$; at the regular ends $p - \langle \alpha q \rangle$ equals $(c+1)(2bk-1)$ for $\sigma = -1$ and $k(a^2 - 2(c+1)b) + (b-c-1)$ for $\sigma = +1$, both exceeding $J-1$ by (F4) ($b \le a < 2b$); and the final end is excluded as in step (4) of Theorem 8.4. So $\eta_J = f = -4Js + R_J$, and the Faulhaber sums give the stated values (certificate, Remark 8.11). For $n \ge 3$ the values are nonzero since $P_{n-1}, P_{n-2} \ge 1$; for $n = 2$, $P_0 = 0$. □

Lemma 8.7. *$\eta_J(L(2J^2-1,\ J-1)) = -2$ for every $J \ge 2$.*

Proof. $p = (J-1)(2J+2)+1$, so $s(J-1,p) = \mathcal{T}(J-1,p) - s(1,J-1)$ and $q^* \equiv -(2J+2)$. No residue wraps, as $\alpha(2J+2) \le p-1$ for $\alpha \le J-1$. The box is $\{\pm(1,-(J-1))\}$, giving $\mathcal{B}_J = -2$, and the bulk $-4Js + R_J$ vanishes identically (certificate, Remark 8.11). □

Lemma 8.8. *$\eta_J(L(2J^2-1,\ 1)) = -J(J-1)(2J-1)/3$ for every $J \ge 2$.*

Proof. $s(1,p) = (p-1)(p-2)/(12p)$ and $\rho_p(\alpha) = p - 2\alpha$. The box is $\{(u,-u) :\ 1 \le |u| \le J-1\}$, since $p > 2J-2$, so $\mathcal{B}_J = -J(J-1)$. Summing gives the value (certificate, Remark 8.11). □

Corollary 8.9 (Rank-uniform zeros of norm ±1 and ±2)**.** *The S-unit orbits of norm ±1 or ±2 are ±1, $\pm\delta^{\pm n}$ and the norm-two orbits $a_nJ \pm P_n$ ($n \ge 0$). Those for which η_J vanishes at every rank $J \ge 2$, or at all sufficiently large ranks, are exactly the first three families of [1, Thm. 7.1]: $q \equiv J$, $J+1$, $2J-1$, up to orbit. The fourth family 2δ has norm -4 and is not covered.*

Proof. Norm ±1. $q = \pm1$ never vanishes (Lemma 8.8); δ is the third family; δ^n with $n \ge 2$ is nonzero on $J \equiv 1 \pmod{P_n}$ (Theorem 6.1(ii)). *Norm ±2,* $n \ge 3$. These orbits are nonzero on $J \equiv 0 \pmod{a_n}$ (Theorem 8.6). $n = 2$, *i.e.* $q = 3J \mp 2$. Both orbits are nonzero on the class $J \equiv 1 \pmod 3$, where the procedure's certified polynomial is $\mathcal{Q}(k) = 2k$ for both signs. $n = 1$. $q = J+1$ is the second family; $q = J-1$ gives -2 (Lemma 8.7). $n = 0$. $q = J$ is the first family. □

Observation 8.10 (Zero classes in the proved sectors)**.** The decision procedure gives the complete zero-class sets below ($J \equiv j$ modulo the period); the entries for δ^{10} and for $n = 8$ in S_1 were never computed before the procedure was fixed. For δ^{10} the 58 classes are enumerated in the supplementary JSON cited after the table; only their number is printed in the table.

orbit	period	zero classes
δ	1	all $J \geq 2$ (third family)
δ^2	2	$\{0\}$ (Theorem 5.1)
δ^3	5	$\{0\}$
δ^4	12	$\{3, 9\}$
δ^5	29	$\{0, 10, 19\}$
δ^6	70	$\{7, 10, 13, 14, 20, 50, 56, 57, 60, 63\}$
δ^7	169	$\{0, 24, 46, 75, 94, 123, 145\}$
δ^8	408	$\{17, 24, 34, 48, 72, 85, 96, 109, 119,$ $289, 299, 312, 323, 336, 360, 374, 384, 391\}$
δ^9	985	$\{0, 193, 792\}$
δ^{10}	2378	58 classes
$a_n J - P_n$ / $a_n J + P_n$, $n = 2$	3	$\{0\}$ / $\{0\}$
$n = 3$	7	$\{2\}$ / $\{5\}$
$n = 4$	17	$\{4, 13, 14\}$ / $\{3, 4, 13\}$
$n = 5$	41	$\{8, 12, 27\}$ / $\{14, 29, 33\}$
$n = 6$	99	$\{20, 21, 28, 31, 71, 78, 79\}$ / $\{20, 21, 28, 68, 71, 78, 79\}$
$n = 7$	239	$\{41, 46, 70, 75, 113, 193\}$ / $\{46, 126, 164, 169, 193, 198\}$
$n = 8$	577	$\{32, 119, 120, 122, 168, 409, 410, 439, 457, 458, 547\}$ / $\{30, 119, 120, 138, 167, 168, 409, 455, 457, 458, 545\}$

The full list of 58 classes for δ^{10} is included in `results/LC14B_ZERO_SETS.json` of the supplement [7]. Across these 24 families the largest threshold k_j^* is 2 and every $\mathcal{Q}_j$ has degree at most 1, as Theorem 8.4 now proves. *Observed, not proved:* the zero sets of δ^n are symmetric under $j \mapsto -j$, and the two signs of the norm-two orbits have mirror-image zero sets.

Remark 8.11 (Symbolic certificates)**.** The Faulhaber-sum evaluations behind Theorem 8.6 (four cases (ν, σ)), the vanishing bulk of Lemma 8.7 and the sum of Lemma 8.8 were certified symbolically by sympy (exact rationals, polynomial remainder modulo $a^2 - 2b^2 - \nu$, no numerical simplification), together with a re-run of the identities of Remark 6.4; seven identity sets, all true. The same identities were independently checked in Wolfram Engine 14.3. Theorem 8.6 in all four cases and Lemmas 8.7 and 8.8 agree in both routes. The certificates of Theorem 6.1 remain those of Remark 6.4, sympy and Wolfram Engine 14.3. The leading-coefficient table of step $(3')$ of Theorem 8.4 was certified separately by sympy and by Wolfram Engine 14.3: each part's k^2 coefficient was computed as $\lim_{k\to\infty}(\text{part})/k^2$ from the exact expressions, with the layer-cake parts as symbolic sums over c; the two routes give identical part values and total 0 in all six cases (S_0, ν) and $(\mathsf{S}_1, \nu, \sigma)$. Every identity is also confirmed by exact evaluation on every row of Appendix A.

Remark 8.12 (Verification)**.** Lemma 8.1 was checked against the literal Δ_J and L_J on 181 random triples (p, q, J) with arbitrary moduli $2 \leq p \leq 150$ and $J \leq 40$, and the mutated weight $J - |u| - |v|$ fails. Theorem 8.6 was checked on 70 exploratory rows ($n \leq 8$, $2 \leq k \leq 6$, both signs) and 82 rows fixed before computation ($n \leq 12$ with $k = 7, 8, 9$ and $13 \leq n \leq 16$ with $k = 1, 2$, both signs), the largest being $n = 16$, $J = 1{,}331{,}714$ with $\eta_J(L(p, a_{16}J - P_{16})) = 25{,}356{,}500{,}416 = \frac{1}{3}P_{16}P_{14} \cdot 2$; also $n = 15$, $J = 551{,}614$: $\eta_J = -13{,}051{,}463{,}048 = -P_{14}^2 \cdot 2$; the sign-flipped statement fails on every odd-n held-out row. Lemmas 8.7 and 8.8 were checked at $J = 401, \ldots, 410$ and at $J = 10{,}000$, resp. $J = 5{,}000$. For Theorem 8.4 the procedure's polynomial $\mathcal{Q}_j$ of every class of the 24 families predicted η_J at $k_j^* + 3$ and $k_j^* + 4$ (and, for δ^n with $n \leq 8$ and the norm-two orbits with $n \leq 6$, at $3k_j^* + 11$) before those rows were computed: 13,080 predictions, none missed. A fit with the wrong period $P_n + 1$ mispredicts 6, 13, 30, 71 and 170 classes for $n = 3, \ldots, 7$.

Corollary 8.13 (Unbounded ambiguity of a vanishing eta invariant)**.** *For every $K \geq 1$ there are a rank J and the level $p = 2J^2 - 1$ with K chiral lens spaces $L(p, q_1), \ldots, L(p, q_K)$ whose orbits $\{\pm q_i^{\pm 1}\}$ are pairwise distinct (so no two are homeomorphic, even after orientation reversal), all with $\eta_J(L(p, q_i)) = 0$ and none rank-J balanced. One can take the q_i to be the reductions of δ^{n_i} for any K distinct odd $n_i \geq 3$, and J any sufficiently large multiple of* $\operatorname{lcm}(P_{n_1}, \ldots, P_{n_K})$.

Proof. Fix distinct odd $n_1, \ldots, n_K \geq 3$, put $\Lambda = \operatorname{lcm}(P_{n_1}, \ldots, P_{n_K})$ and let J be a multiple of Λ. Then $P_{n_i} \mid J$ and Theorem 6.1(i) gives $\eta_J(L(p, q_{n_i})) = 0$ for every i. Since (5) is a ring homomorphism, $q_n q_m \equiv q_{n+m} \pmod p$. Hence each of the relations $q_{n_i} \equiv \pm q_{n_j}$ $(i \neq j)$ and $q_{n_i} q_{n_j} \equiv \pm 1$ (the orbit collisions), $q_{n_i}^2 \equiv -1$ (the mirror condition, i.e. $q_{2n_i} + 1 \equiv 0$) and $q_{n_i} \equiv \pm(2J \pm 1)$ (family (b) of Theorem 2.2) says that p divides one of the integers

$$a_{n_i} \mp a_{n_j} + 2(P_{n_i} \mp P_{n_j})J, \qquad a_{n_i+n_j} \mp 1 + 2P_{n_i+n_j}J, \qquad a_{2n_i} + 1 + 2P_{2n_i}J, \qquad a_{n_i} \mp 1 + 2(P_{n_i} \mp 1)J,$$

each an affine function of J whose coefficient of J is nonzero (P_n is strictly increasing and $P_{n_i} \geq 5$). Finitely many such functions occur; each is nonzero and of absolute value less than $p = 2J^2 - 1$ for all large J, so p divides none of them. Family (c) of Theorem 2.2 is impossible because $p > 2J$. Hence, for all sufficiently large multiples J of Λ, the K orbits are pairwise distinct, none is mirror, and by Theorem 2.2 none of the $L(p, q_{n_i})$ is rank-J balanced. □

This deduction from Theorem 6.1 and Theorem 2.2 says that the single number η_J can lose an unbounded amount of information about the handedness of the geometry: at a suitable rank, as many pairwise non-homeomorphic chiral lens spaces as desired share the reading $\eta_J = 0$ while having unequal level counts. It says nothing about the full level sequences $(D_J(n))_{n \geq 0}$, the all-rank sequences $(\eta_J)_{J \geq 1}$ or other invariants, which may well separate them.

Remark 8.14 (Rank one against higher rank)**.** At $J = 1$ the opposite holds: $\eta_1(L(p, q)) = -4s(q, p)$ vanishes iff $q^2 \equiv -1 \pmod p$ [6, p. 36], iff $L(p, q)$ is amphicheiral, and then it is balanced at every level [2, Cor. 4.4]; so a zero of η_1 is exactly a mirror symmetry. (As an independent check of the convention, $\eta_1 = 0 \iff q^2 \equiv -1$ was re-verified with the literal Dedekind sum for every $2 \leq p < 400$ and every unit q: 191 zeros among 48,517 pairs, no exception; Appendix A.) At prime moduli $p \geq 5$ the dichotomy survives at every rank with $2J^2 \not\equiv 1 \pmod p$ and $p \nmid J$ (Corollary 2.6(c)); at the blind ranks $p \mid J$ every $q \not\equiv \pm 1$ is a zero [2, Cor. 6.4(d)]. Corollary 8.13 shows that at silver ranks it fails without bound. In the language of chemistry, the chiral spaces of Corollary 8.13 are cryptochiral for η_J: handed, with a vanishing net helicity.

9 Open problems

(1) *The period law beyond norm* ± 2. Prove Conjecture 7.1 for the orbits with $|m| \geq 2$, for example the fourth family $2\delta = (1, 2)$ and the orbit $(1, -3)$ of $q \equiv 4 - 4J$, with effective onsets.

(2) *Symmetries.* The reflection $j \mapsto -j$ of the zero classes of δ^n and the mirror relation between the two signs of the norm-two orbits are observed only.

(3) *Zeros outside the S-units.* At $J = 20, 30, 40$ there are about $6J$ zeros per rank, most of them with an empty box (Observation 7.3(b)); they are not classified.

(4) *Divisors.* At divisors $p < 2J^2 - 1$ of $2J^2 - 1$ only the universal orbits of Observation 4.4(a) are proved to be zeros; Sections 7 and 8 concern the level $p = 2J^2 - 1$ only.

(5) *A finite description.* Whether the full zero set at $p = 2J^2 - 1$ admits a finite parametrized description is open (Observation 7.3(c)).

(6) *Literature.* The Dedekind-sum evaluations at Pell-type arguments used here, and Corollary 2.8, are likely classical; no literature search has been made for them.

A Exact computations

Every statement was accompanied by exact computations in integer or rational arithmetic, several with predictions fixed before the corresponding rows were computed ("held-out"; "exploratory" rows were available when the statement was formulated). They test the identities and conventions; they do not replace the proofs. The independent evaluator named below is the pinned exact implementation of the master formula used throughout the companion [2].

statement	what was checked	rows	failures
Remark 8.14	$s(q,p) = 0 \iff q^2 \equiv -1 \pmod p$ by the literal Dedekind sum, every $2 \le p < 400$, every unit q (direct Dedekind sums)	48,517 (191 zeros)	0
Observation 4.4(a), Definition 4.2	re-run of the exploration searches, ranges recovered from the retained scripts and the outcome predicted before computation: the 630 S-units $s \cdot 2^c \sqrt{2}^{\,a} \delta^b$ ($\lvert c\rvert \le 2$, $\lvert a\rvert \le 3$, $\lvert b\rvert \le 4$) and $\rho = \pm\delta^k$ ($k \le 14$) at the 57 silver primes $17 \le p \le 673$, pinned exact evaluator; universal sets exactly as predicted; the named orbits lie in D_p for every divisor $p > 1$ of $2J^2 - 1$, $J \le 100$; counterexamples at $p = 49$	57 primes; 1,734 values	0
Theorem 5.1 (conjectural stage)	the value $(J-1)\mathbf{1}_{J \text{ odd}}$ predicted before computation on every $J \in [19, 60]$ with $p = 2J^2 - 1$ prime; the anchors $p = 17, 31, 97, 127$ re-verified; distinctness of the orbit of $3 - 4J$ from the four orbits of Lemma 4.1, family (b), the orbit of Theorem 2.7 and the mirror orbit on every row; the inverse orbit $3 + 4J$ gives the same value on every row	19 (held-out)	0
Conjecture 4.6	$\eta_J(L(p, 3 - 4J)) \ne 0$ predicted for every prime $p \equiv 1 \pmod 8$ with $761 \le p \le 1553$ at its minimal silver rank	28 (held-out)	1 ($p = 1129$, $J = 203$: $\eta = 0$)

statement	what was checked	rows	failures
Theorem 5.1	each of s, R_J, Δ_J, L_J against its closed form (direct Dedekind sum, direct residues, enumeration of the triangle, enumeration of $W \times W$), the assembled value against (9), and against the independent evaluator, on every $J \le 60$ with p prime (31 exploratory rows), every $2 \le J \le 80$ with p composite (41) and every $61 \le J \le 100$ with p prime (14); controls: dropping L_J changes every odd-J row; the inverse orbit gives the same value (18 rows); the control premise "$q = 4 - 4J$ is nonzero on some even-J row" failed ($\eta = 0$ on all 8 even rows checked, Remark 5.2(iii))	86 (55 held-out, 41 composite)	0 (one control premise failed)
Theorem 6.1, Lemma 6.3	on every row: the unit identity $q_n(-\nu_n t) \equiv 1$; $s(q_n, p)$ by a generic reciprocity algorithm against (18), (20) (against the literal Dedekind sum for $J \le 120$; against Corollary 2.8 on every $n = 2$ row); Lemma 6.3; R_J, Δ_J, L_J from their definitions against (19)–(23) (against the literal functions of the previous row for $J \le 100$); η_J against the claim (against the pinned evaluator for $J \le 60$). Rows: slice (i) $n \le 10$, $k \le 6$ (59, exploratory) and 42 rows never computed before with $n \le 16$; slice (ii) $n \le 10$, $k \le 8$ (72, exploratory) and 39 never computed before with $n \le 16$; largest $J = 941{,}664$ (slice (i)) and 941,665 (slice (ii)). Faulhaber-sum identities (Remark 6.4): four identity sets by sympy and by Wolfram Engine 14.3. Fast evaluator against the literal terms and the pinned evaluator: 106 rows. Controls: $R_J - 4Js \ne \eta_J$ on all 39 held-out slice-(ii) rows; $\eta(q^*) = \eta(q)$, $\eta(-q) = -\eta(q)$ on the 65 held-out rows with $n \le 12$; the mutated modulus a_n refuted at $n = 3, 5, 7$ (9 fresh rows)	212 (81 held-out); 8; 106; 9	0
Conjecture 7.1	one polynomial per class modulo $\lvert y \rvert$ fitted on $J \in [150, 400]$ for each of the 64 orbits of $\mathcal{W}$ (no fit failure); 12,800 values at $J \in [1401, 1600]$ predicted from the fits before computation (summary counts retained, not the individual values); on these ranges the identically vanishing orbits are exactly $(0, 1)$, $(1, -1)$, $(1, 0)$, $(1, 2)$	12,800 (held-out)	0
Corollary 7.2(a)	one exact nonzero value η_J at a rank $J \le 6$ for each of the 60 orbits of $\mathcal{W}$ outside the four families, each cross-checked against the pinned independent evaluator (ledger printed in Section 7)	60	0

statement	what was checked	rows	failures
Lemma 8.1	the box sum against the literal Δ_J and L_J on random (p, q, J) with $2 \le p \le 150$, $J \le 40$ (draws with $\gcd(q, p) > 1$ skipped); control: the weight $J - \lvert u\rvert - \lvert v\rvert$ fails	181 (held-out)	0
Theorem 8.6	on every row: $q \cdot 2\nu_n t \equiv 1$, empty box, closed form of s, η_J against the claim; $n \le 8$, $2 \le k \le 6$, both signs (exploratory), and $n \le 12$ with $k \in \{7, 8, 9\}$, $13 \le n \le 16$ with $k \in \{1, 2\}$, both signs (held-out; largest $J = 1{,}331{,}714$); control: the sign-flipped claim fails on every odd-n held-out row	70; 82 (held-out)	0
Lemmas 8.7, 8.8	$J = 401, \ldots, 410$ and 10,000 (Lemma 8.7); $J = 401, \ldots, 410$ and 5,000 (Lemma 8.8)	22 (held-out)	0
Theorem 8.4 (decision procedure)	thresholds and per-class polynomials for the 24 families δ^n, $n \le 10$, and $a_n J \pm P_n$, $n \le 8$ (δ^{10} and $n = 8$ never computed before); every class predicted at $k_j^* + 3$, $k_j^* + 4$ (and $3k_j^* + 11$ for δ^n, $n \le 8$, and $a_n J \pm P_n$, $n \le 6$) before computation; control: period $P_n + 1$ mispredicts $6, 13, 30, 71, 170$ classes for $n = 3, \ldots, 7$; symbolic certificates by sympy and by Wolfram Engine 14.3 (Remark 8.11)	13,080 (held-out); 7	0
Theorem 8.4 (complete zero sets)	for each class of the 24 families: the class polynomial from three exact values at the threshold; its integer roots past the threshold (exact integer square roots, with a pinned self-test on a root of size 10^{18}), each re-evaluated exactly; every rank below the threshold evaluated exactly; the active breakpoints $\le J + 1$ on every class $j < 40$, $k \le 5$; box parity. Result: one isolated zero ($J = 429$), 363 zeros below the thresholds, every class polynomial of degree ≤ 1	24 families	0
Theorem 8.4 (degree ≤ 1)	the k^2 coefficient of each of the four parts of the bulk (Dedekind, constant, layer cake, α-part) as $\lim(\text{part})/k^2$ from exact expressions, in S_0 ($\nu = \pm 1$) and S_1 ($\nu, \sigma = \pm 1$), by sympy and by Wolfram Engine 14.3; corroboration: the 6,025 retained class triples of the decision-procedure run all have degree ≤ 1	6 cases; 6,025	0

Negative controls (a wrong sign, a dropped term, a wrong modulus, a wrong period) were evaluated in each block and failed at the stated counterexamples. Two nonvanishing predictions were refuted and are recorded above: Conjecture 4.6 at $p = 1129$, and the control premise of Theorem 5.1 at $q = 4 - 4J$. The supplementary integer-root routine uses exact integer arithmetic and includes a regression test above 10^{18}; all retained class polynomials here are linear.

Scope of the computations. The accompanying supplement [7] supplies exact evaluators and retained calculation records for the finite checks summarized above. These checks cover the stated

parameter ranges; the universal statements rest on the written proofs. The counterexamples to Conjecture 4.6 and to the $q = 4 - 4J$ control premise remain part of the reported findings.

Authorship, assistance and rights

Luqman Mahmood directed this work and is responsible for its claims and final text. OpenAI Codex and Anthropic Claude assisted with mathematical development, computation, checking and editing. Internal and automated checks are not external peer review. Computational tests do not replace proofs of universal statements, and no physical identification or experiment is implied. The paper and mathematical data are licensed CC BY 4.0. Software source and executable mathematical verification scripts are excluded from that grant; their rights are reserved unless explicitly stated otherwise.

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